

Workshop 00: Create a Buck Converter with the Average Switch Model

Starting Files

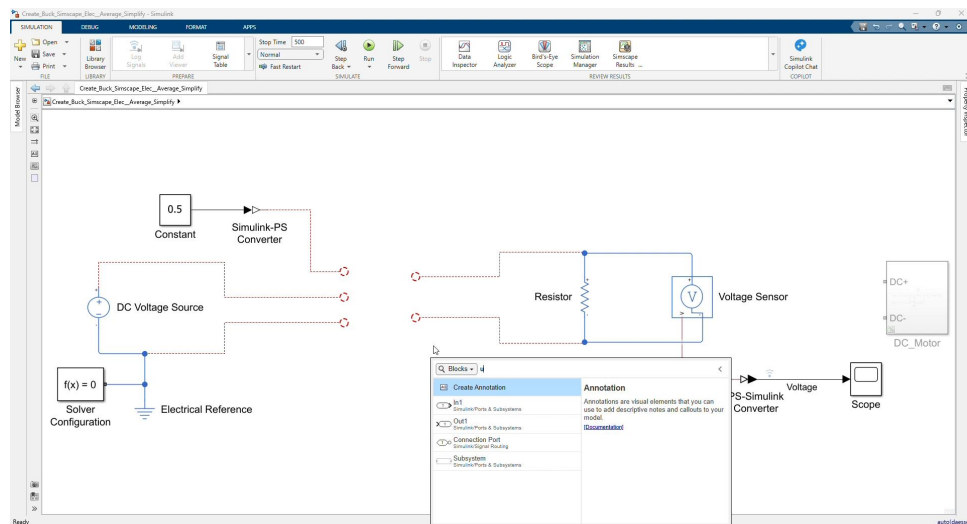
File	Purpose
Create_Buck_Simscape_Elec_Average_Simplify.slx	Starting model

Finished Model Reference

File	Purpose
Create_Buck_Simscape_Elec_Average_Simplify_finished.slx	Finished average buck converter with DC motor load

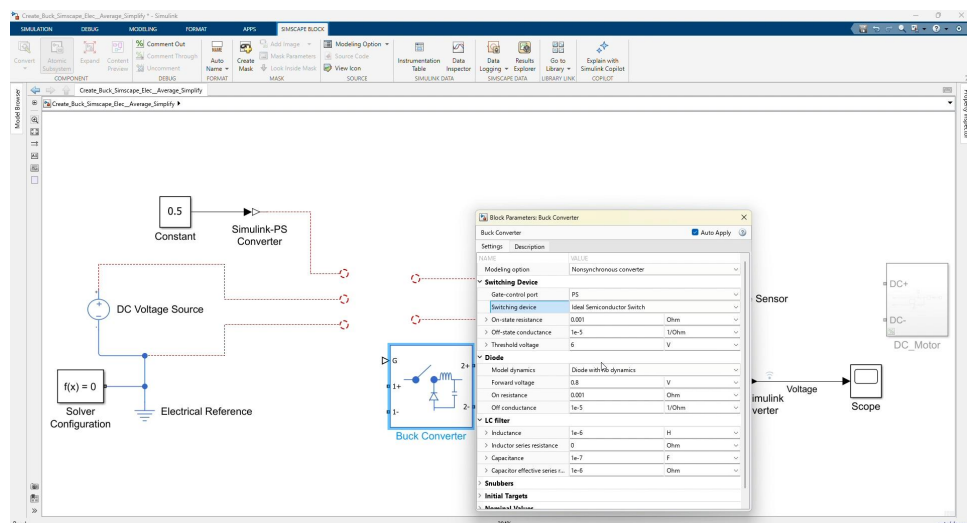
Workshop Procedure

1. Open the starting average buck model and review the prepared source, duty command, resistor load, voltage sensor, and scope.



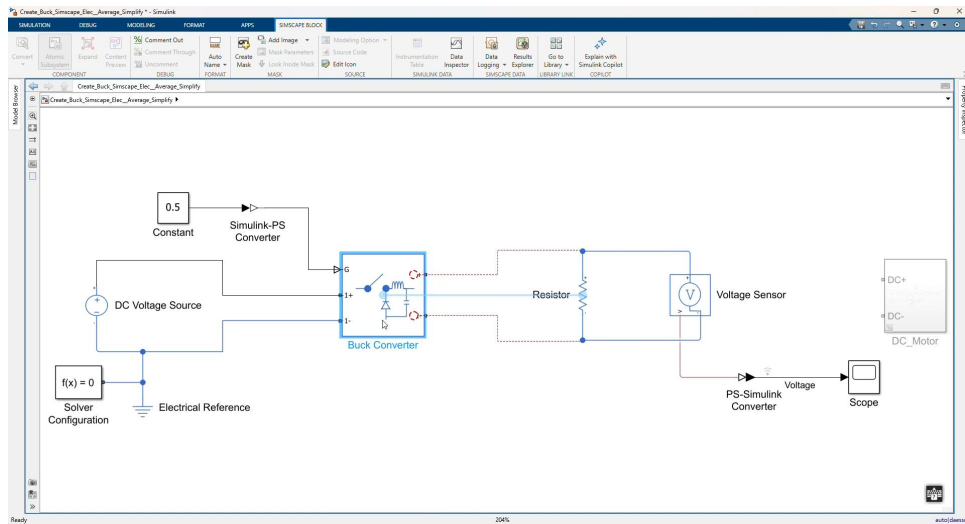
Starting average buck model

2. Add the Simscape Buck Converter block.



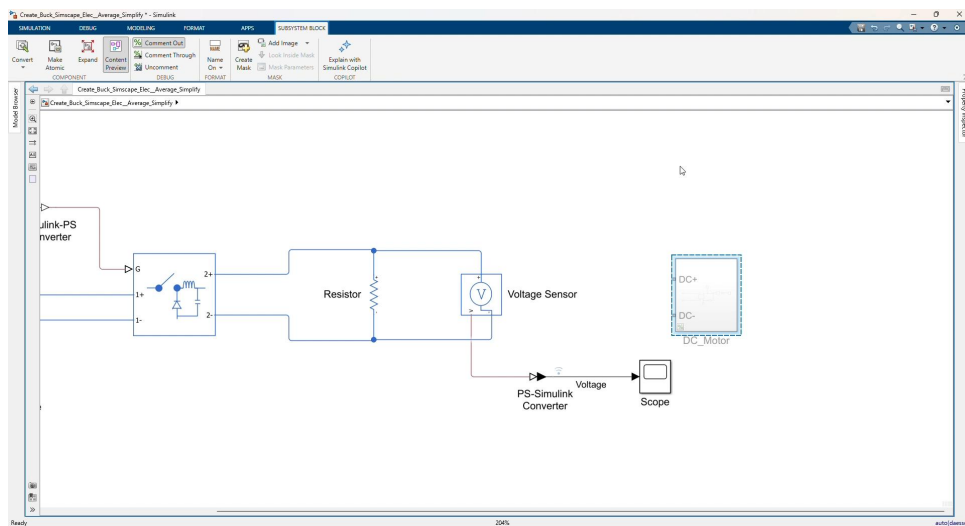
Add Buck Converter block

3. Configure the Buck Converter block as an averaged switching device.



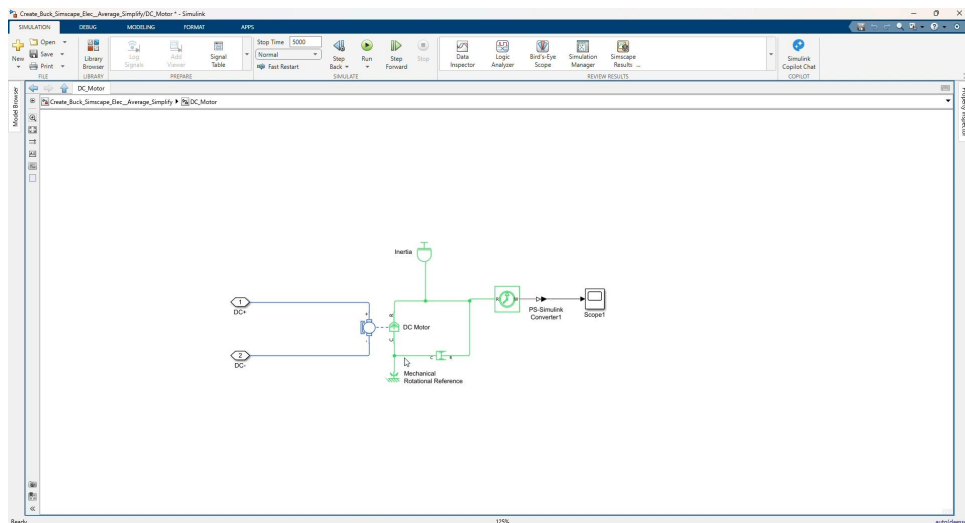
Configure averaged Buck Converter

4. Connect the DC source to the converter input.
5. Connect Constant (0.5) through the Simulink-PS Converter to the converter G input.
6. Connect the resistor load and voltage sensor to the converter output.
7. Run the resistor-load simulation and confirm the output voltage is about 25 V.
8. Create or add the DC_Motor subsystem.

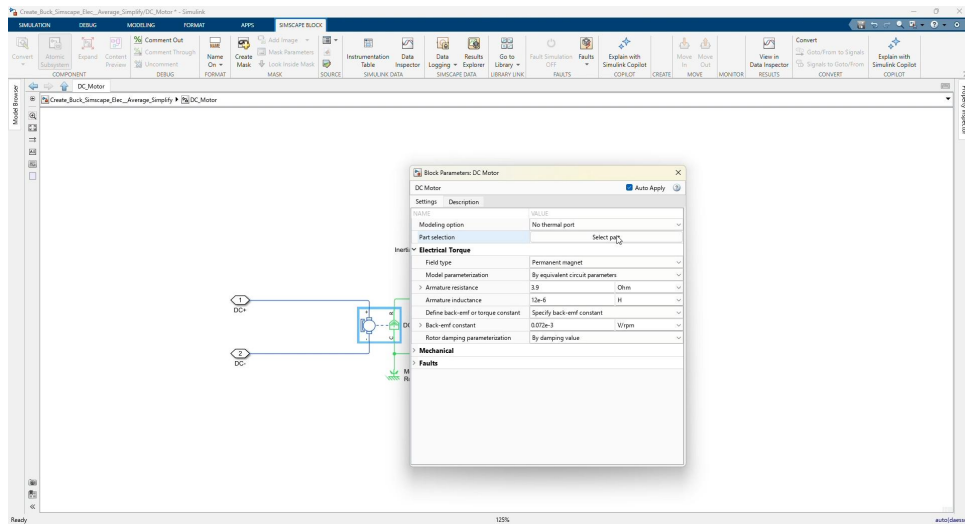


Add DC_Motor subsystem

9. Configure the DC motor, inertia, damper, speed sensor, and speed scope.

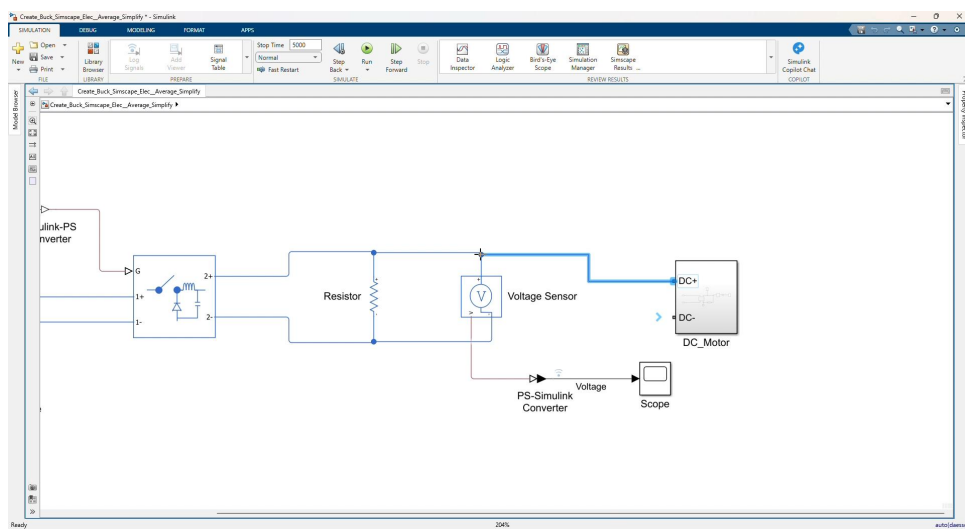


DC motor subsystem



Configure DC motor

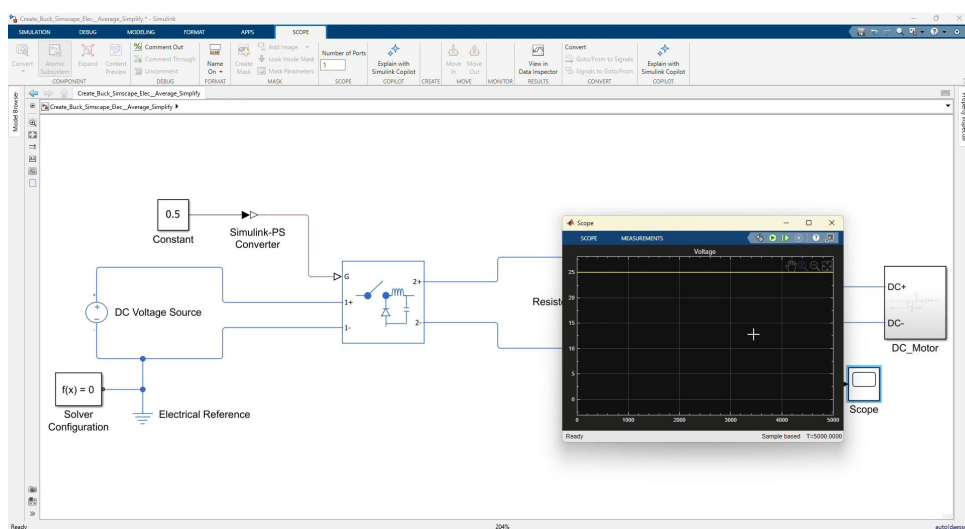
10. Connect DC_Motor in parallel with the resistor load.



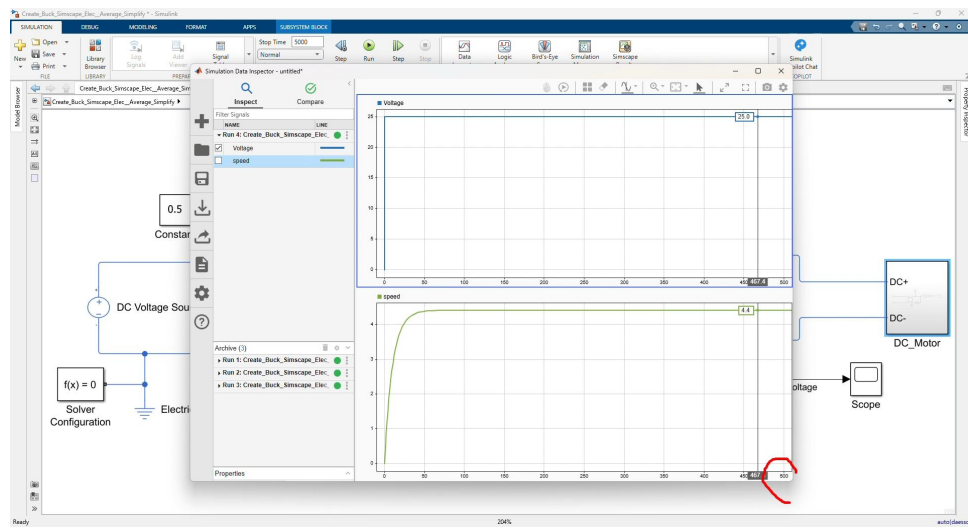
Connect DC_Motor load

11. Set stop time to 5000.

12. Run the final simulation and compare output voltage and motor speed.



Final scope result



Final voltage and speed result

Workshop 01: Create a Simplified Buck Converter with Simscape Electrical

Starting Files

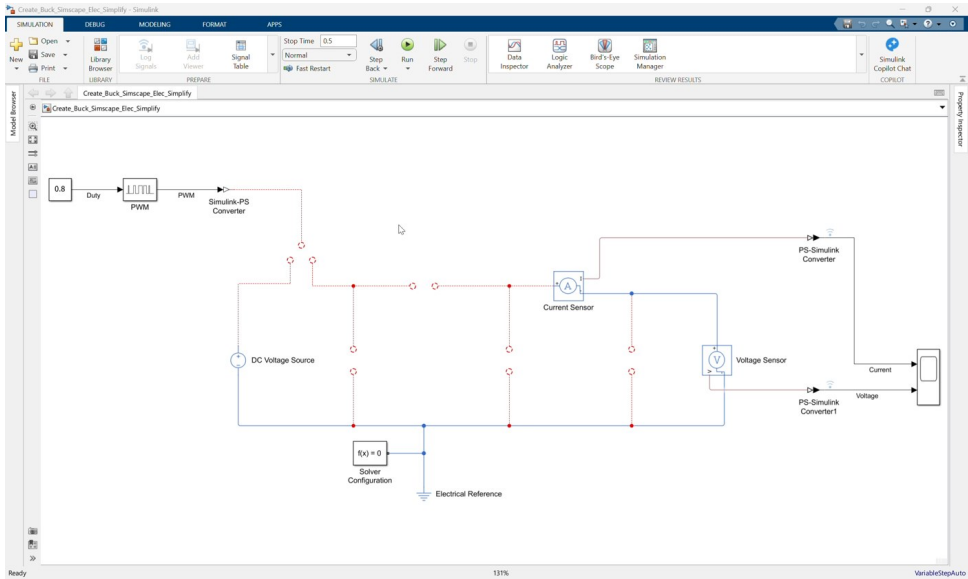
File	Purpose
Create_Buck_Simscape_Elec_Simplify.slx	Starting simplified switching buck converter model

Finished Model Reference

File	Purpose
Create_Buck_Simscape_Elec_Simplify_finished.slx	Finished switching buck converter

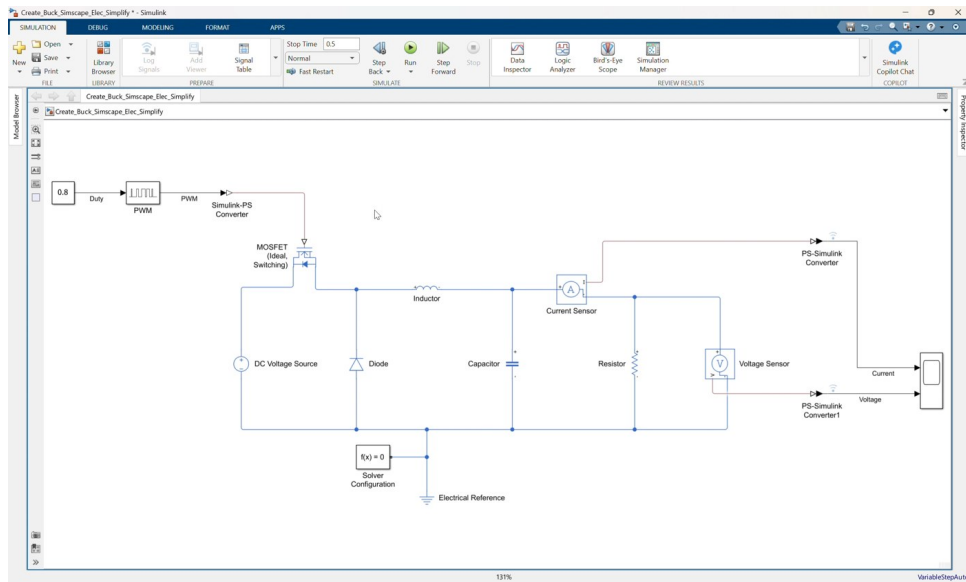
Workshop Procedure

1. Open and review the starting model.



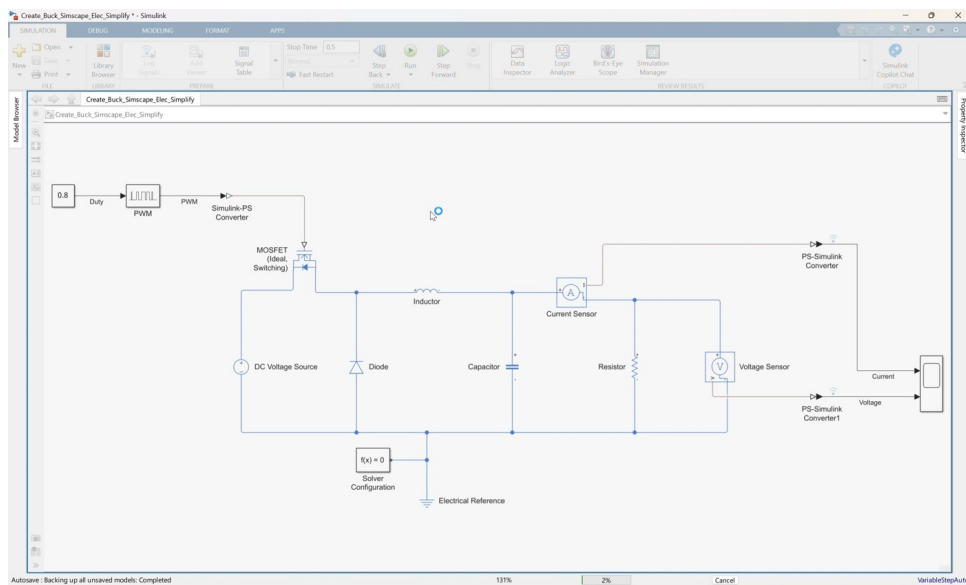
Starting simplified model

2. Add the MOSFET switch.
3. Add the freewheeling diode.
4. Add the inductor.
5. Add the output capacitor.
6. Add the load resistor.
7. Complete the current and voltage measurement connections.



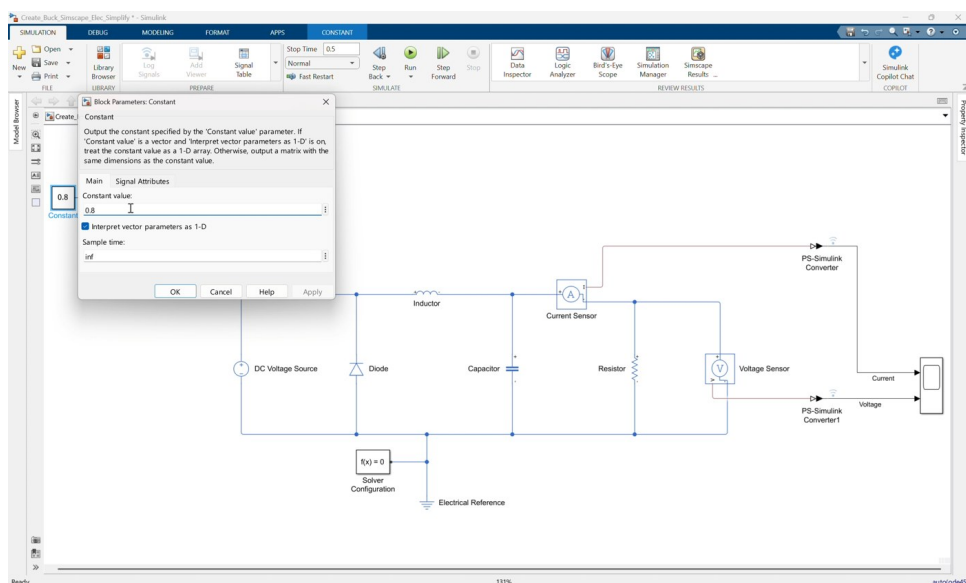
Measurement connections

8. Check the Simscape infrastructure: electrical reference and solver configuration.



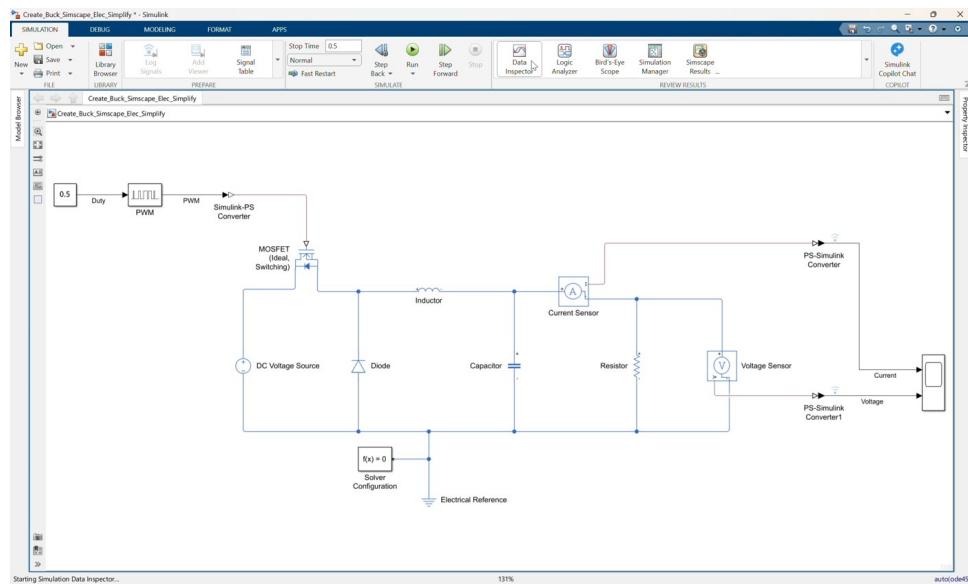
Simscape infrastructure check

9. Set the duty-cycle command. The lesson starts from duty 0.8, then changes to 0.5 near the end.



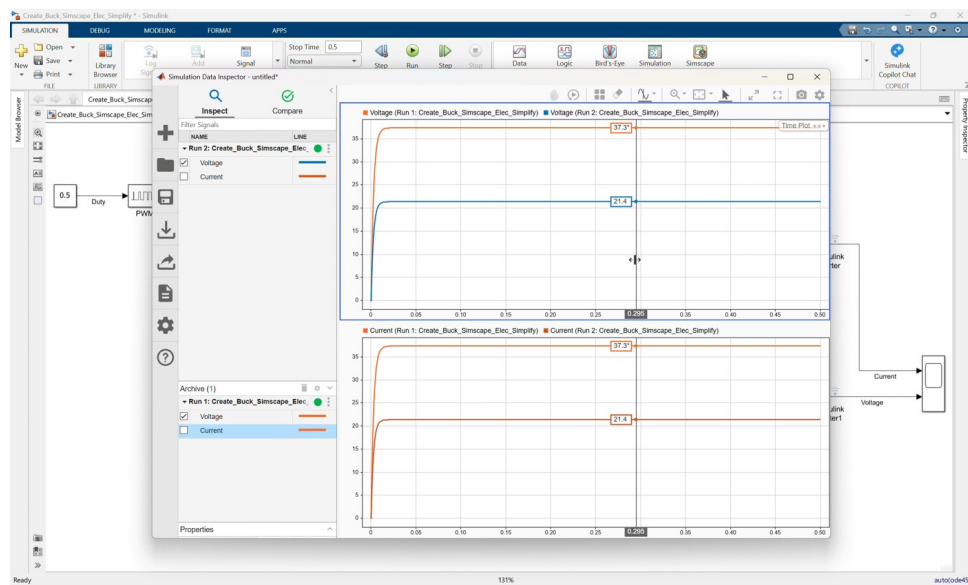
Duty-cycle command change

10. Configure and save the completed model.



Configured completed model

11. Simulate and inspect voltage/current results in the Scope.



Scope result

Workshop 02: Add MOSFET Thermal Behavior to the Simplified Buck Converter

Starting Files

File	Purpose
Create_Temp_Simscape_Elec_Simplify.slx	Starting thermal buck converter model
Thermal_Mosfet_evaluation.slx	Evaluation model for clearer thermal comparison

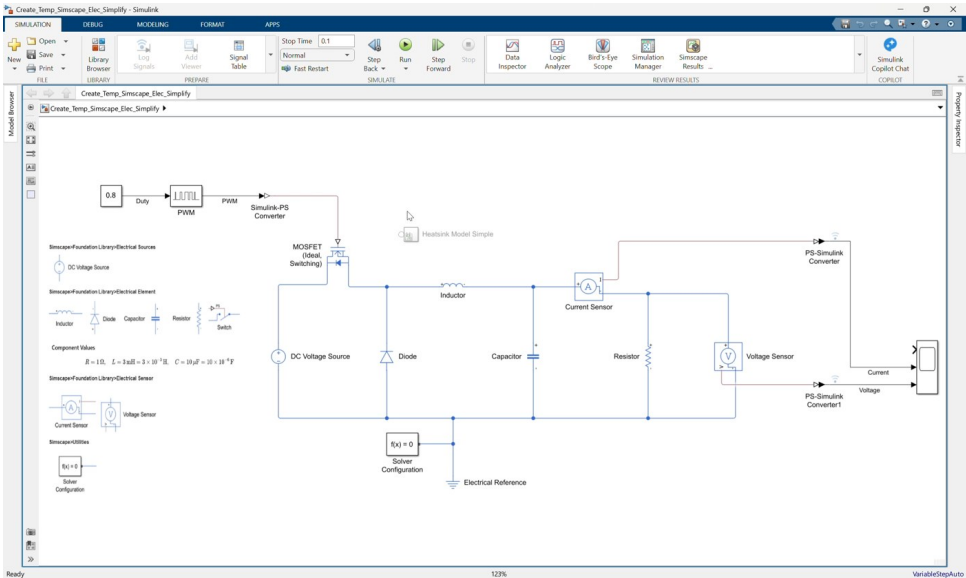
Finished Model Reference

File	Purpose
Create_Temp_Simscape_Elec_Simplify_finished.slx	Finished thermal buck converter model
Thermal_Mosfet_evaluation.slx	Used for the second thermal comparison session

Workshop Procedure

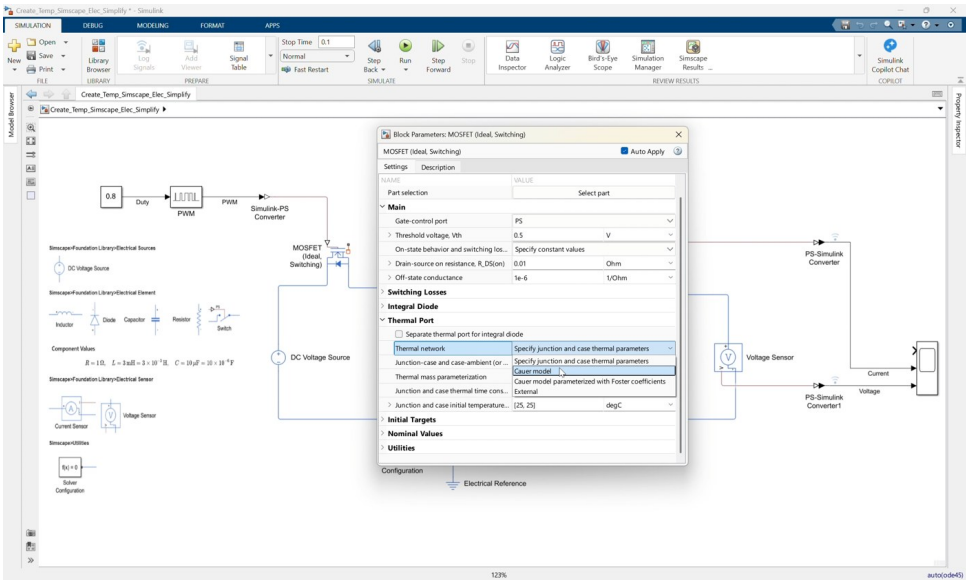
Session 1: Thermal Buck Converter

1. Open and review the starting thermal buck converter model.



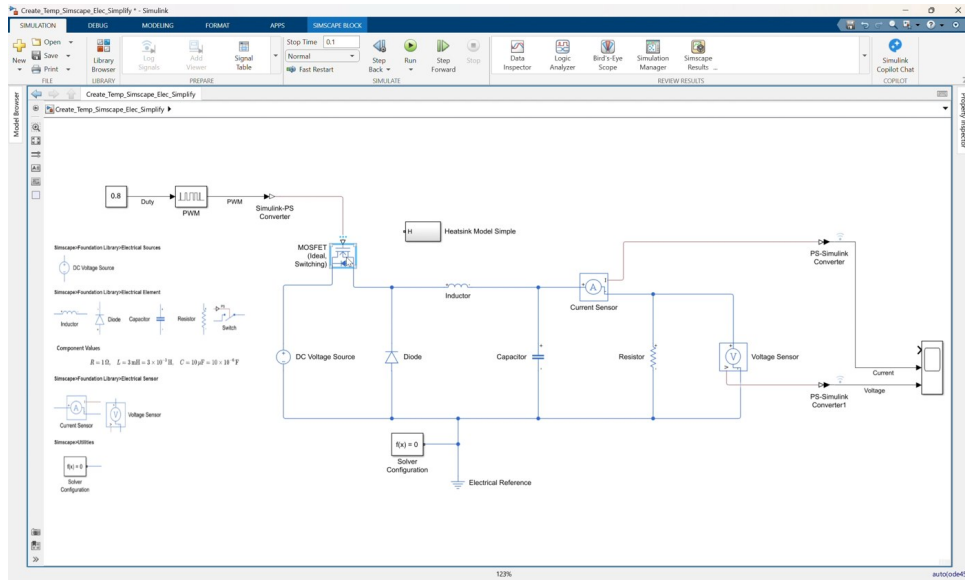
Starting thermal buck converter model

2. Enable the MOSFET thermal port.



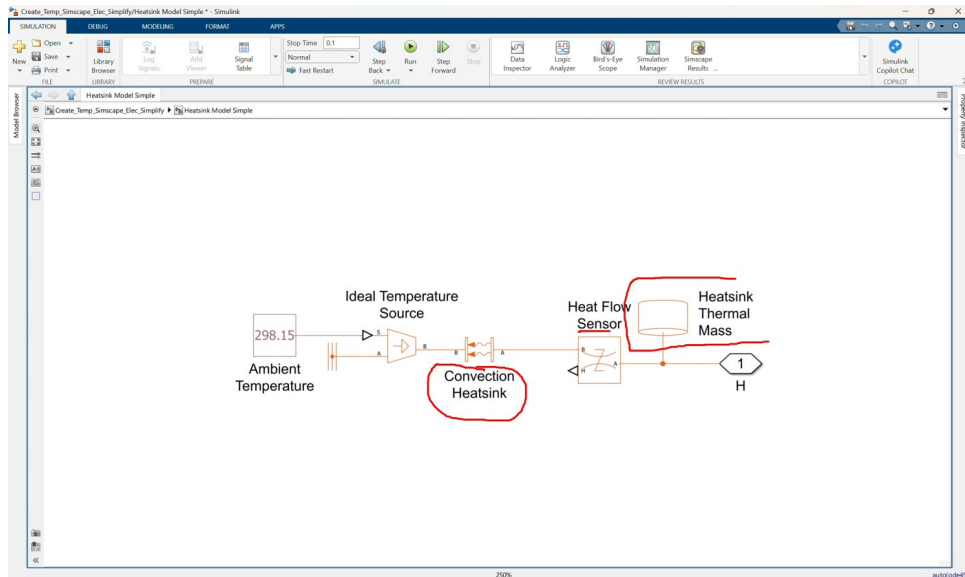
MOSFET thermal port settings

3. Add the heatsink subsystem.



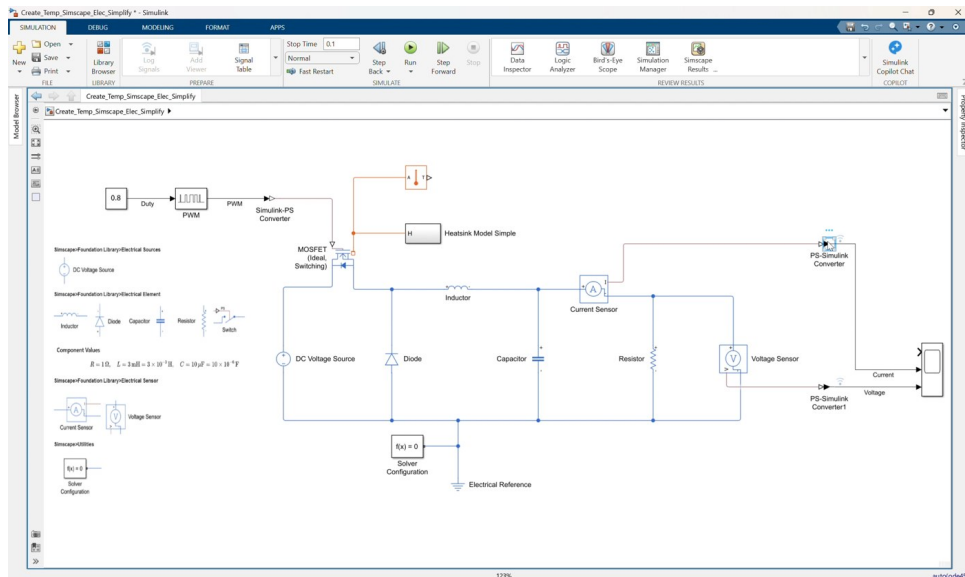
Heatsink subsystem added

4. Review the heatsink subsystem.



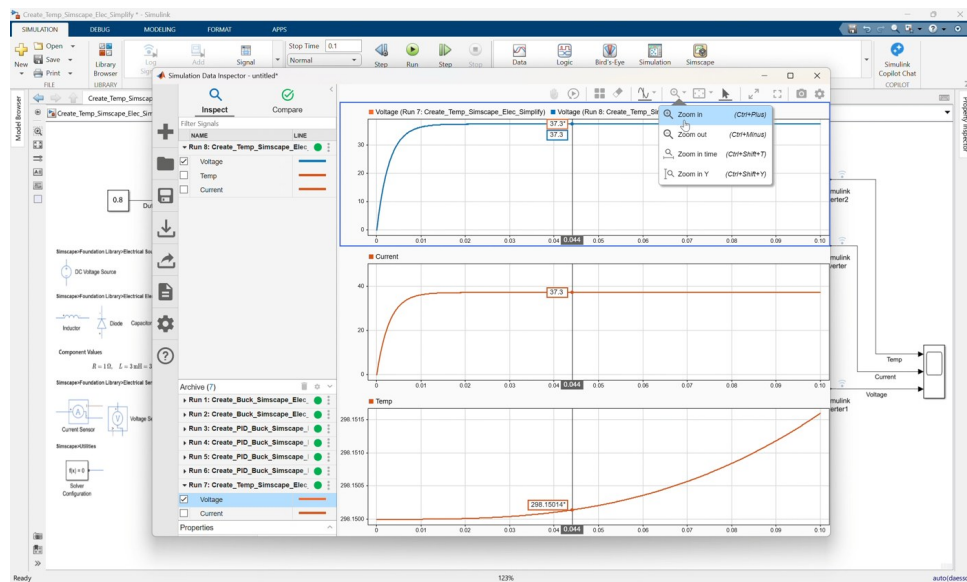
Heatsink subsystem internals

5. Add MOSFET temperature measurement.



Temperature measurement connection

6. Configure and run the thermal buck converter.

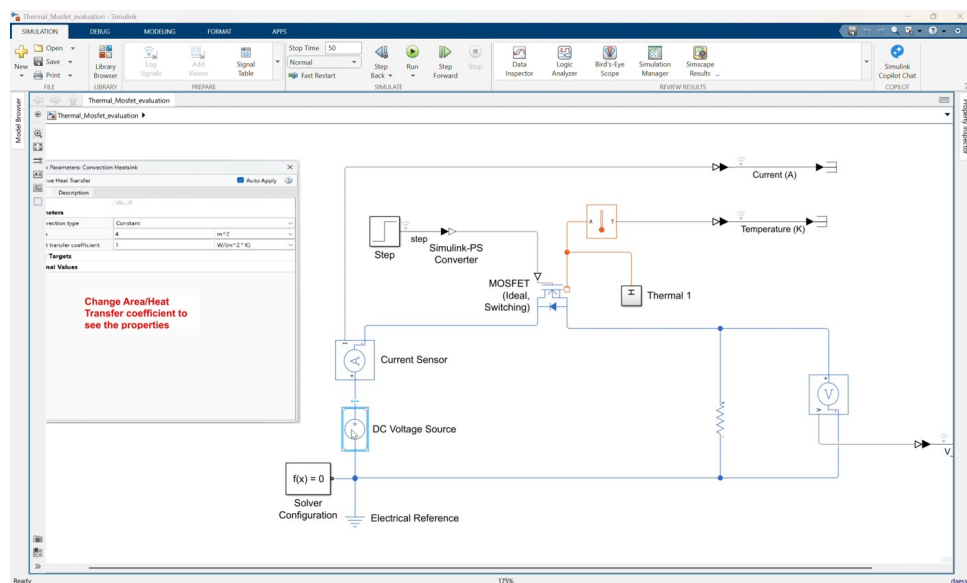


Thermal buck converter result

7. Explain the short-duration Session 1 thermal result.

Session 2: Thermal Evaluation Model

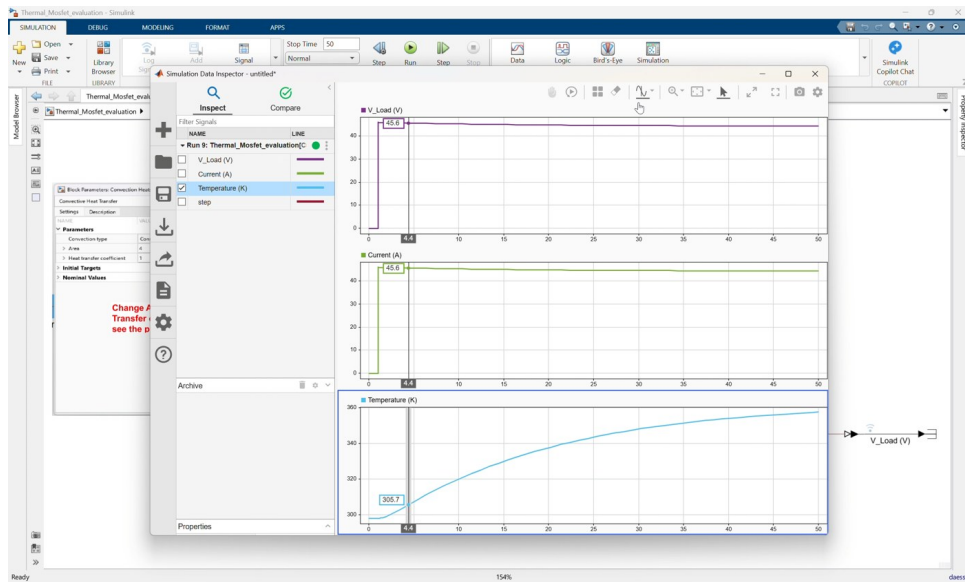
1. Open and review the evaluation model.



Thermal evaluation model

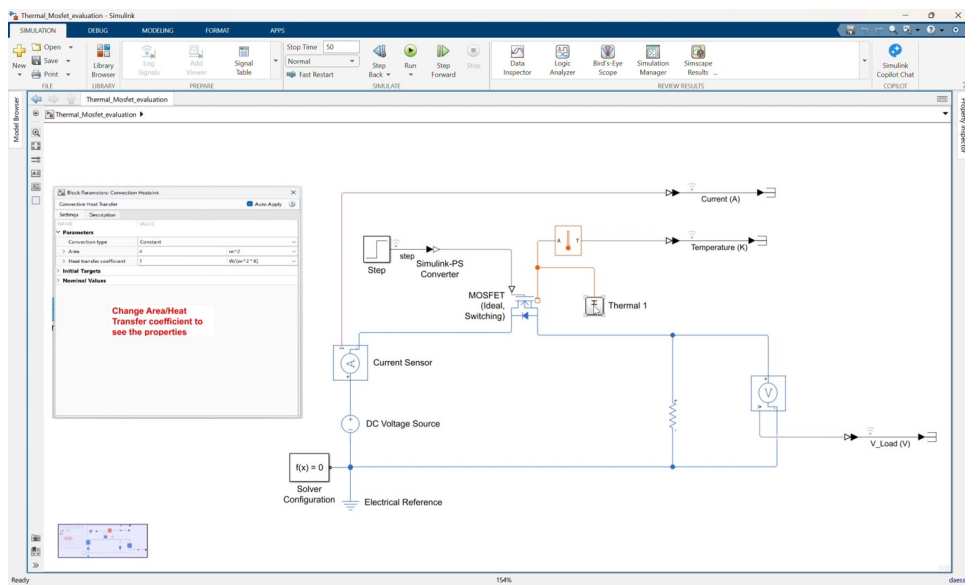
2. Inspect the thermal subsystem and baseline convection settings.

3. Run the baseline case.



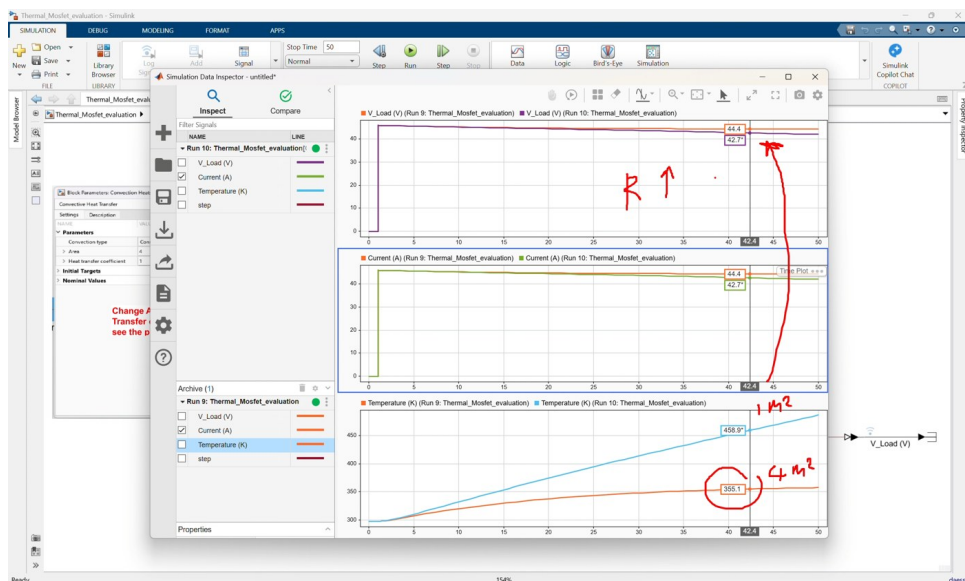
Baseline thermal case

4. Increase convection area from 1 m² to 4 m².



Change convection area to 4 square meters

5. Compare the two thermal cases.



Comparison between 1 and 4 square meter convection area

Workshop 03: Add a PI Controller to the Simplified Buck Converter

Starting Files

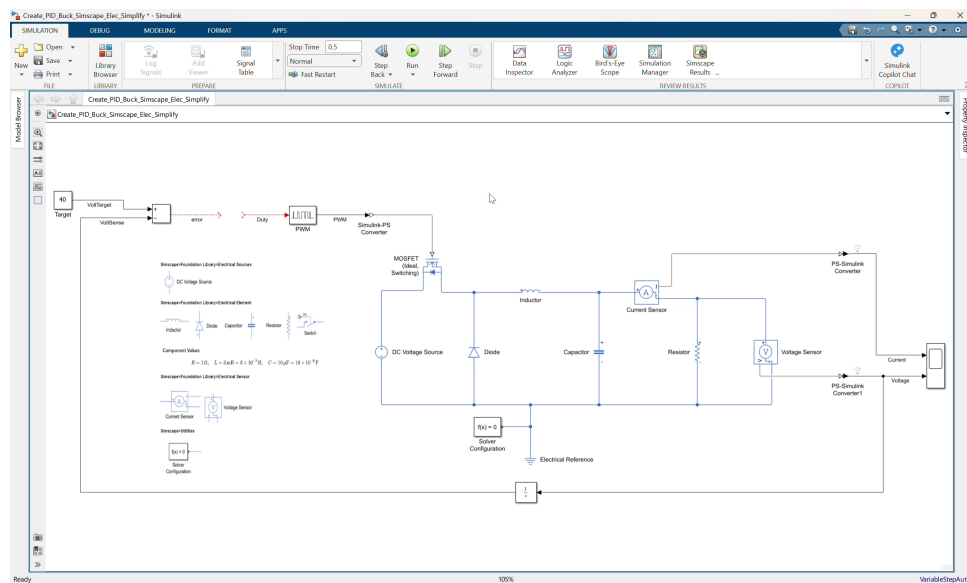
File	Purpose
Create_PID_Buck_Simscape_Elec_Simplify.slx	Starting closed-loop controller model

Finished Model Reference

File	Purpose
Create_PID_Buck_Simscape_Elec_Simplify_finished.slx	Finished PI-controlled buck converter

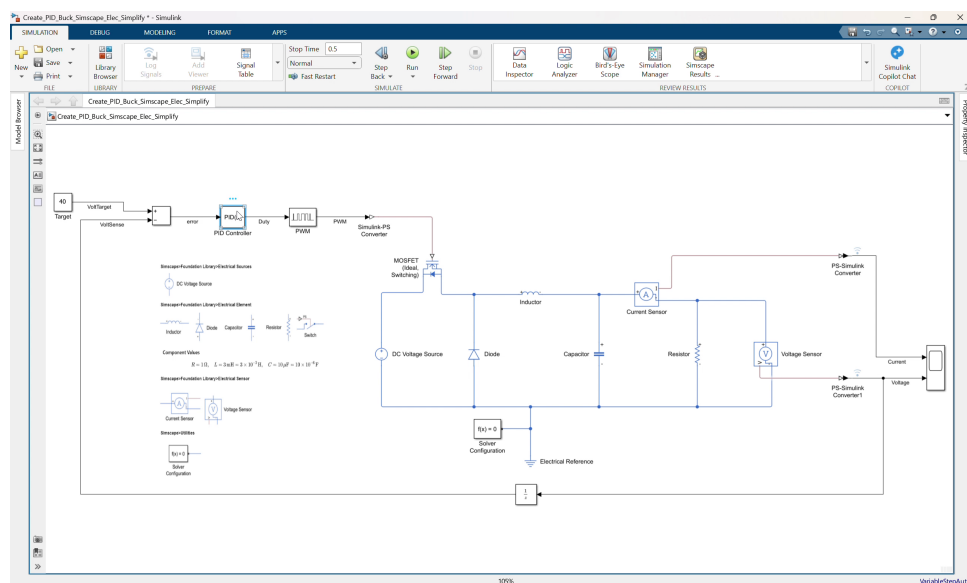
Workshop Procedure

1. Open and review the prepared PID model.



Prepared PID model

2. Explain the closed-loop path: target voltage, sensed voltage, error, controller duty, PWM, and plant.
3. Add the PID Controller block.

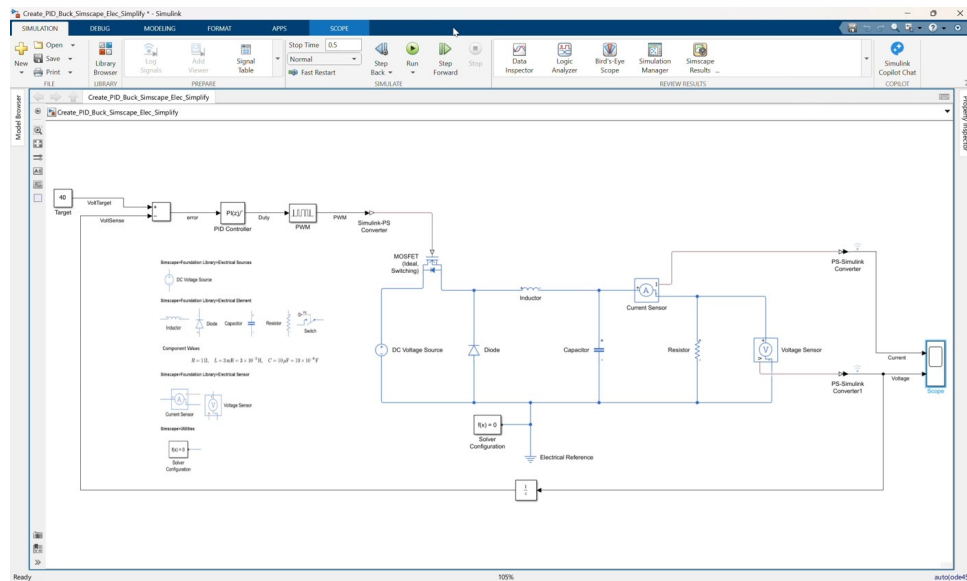


PID Controller block added

4. Configure the controller as a PI controller.
5. Enter initial PI settings.

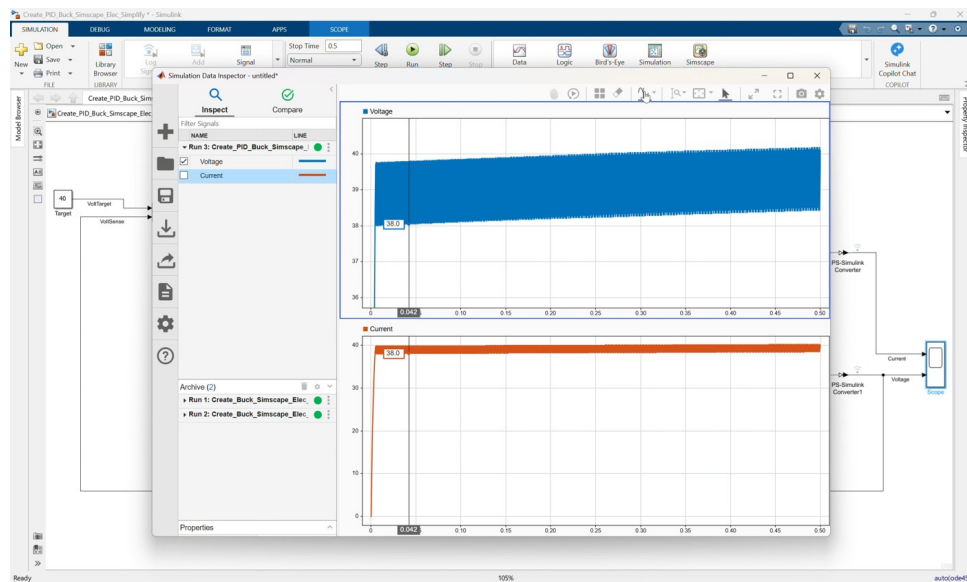
6. Enable output saturation from 0 to 1.

7. Connect the controller between the error signal and PWM duty input.



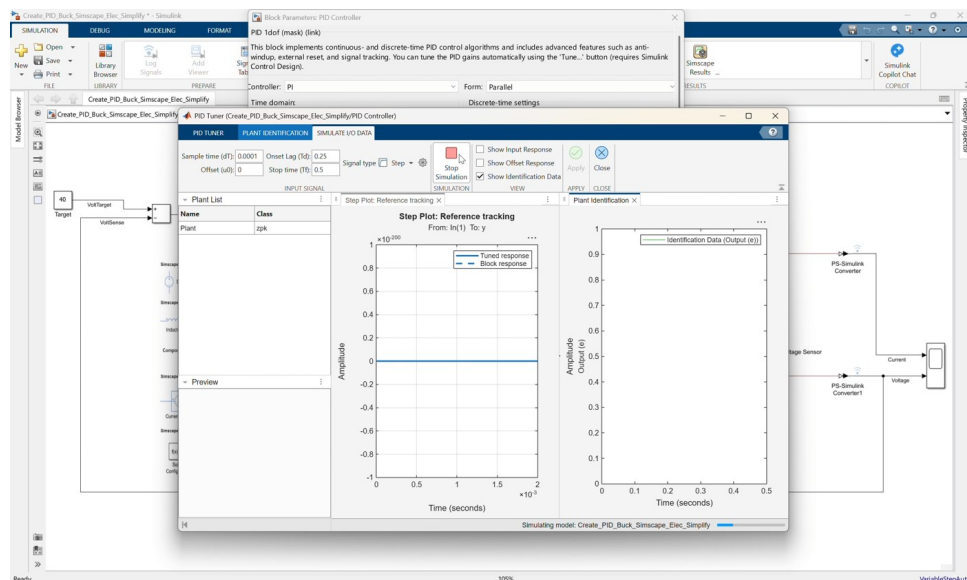
Controller connected to PWM

8. Check the Scope signals.



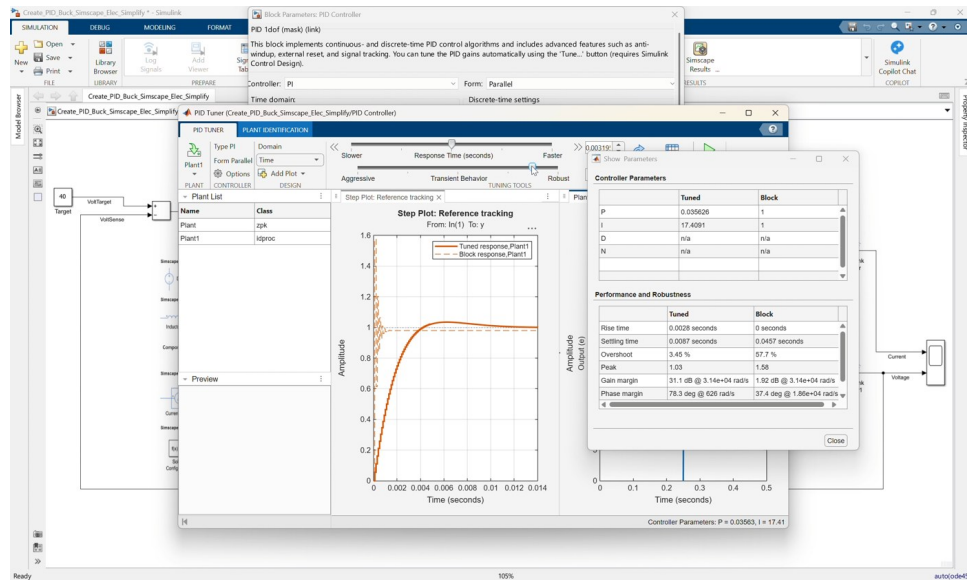
Scope signal check

9. Open PID Tuner and estimate a new plant by linearizing the model at the operating point.



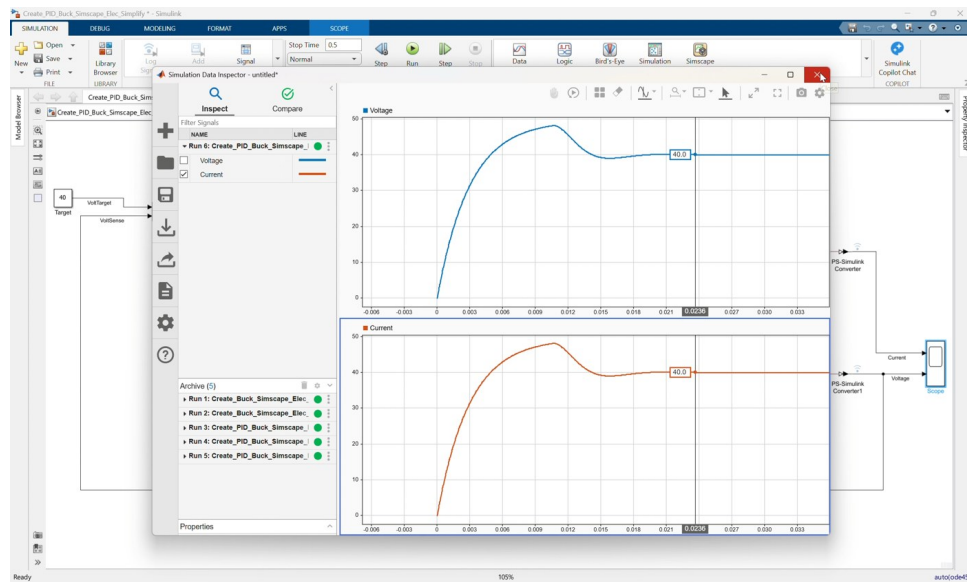
PID Tuner plant estimation

10. Tune the PI response using the PID Tuner response/transient sliders.



PID Tuner slider tuning

11. Apply the tuned gains, simulate, and save the model.



Final tuned response

Workshop 04: Optimize PI Gains for the Simplified Buck Converter

Starting Files

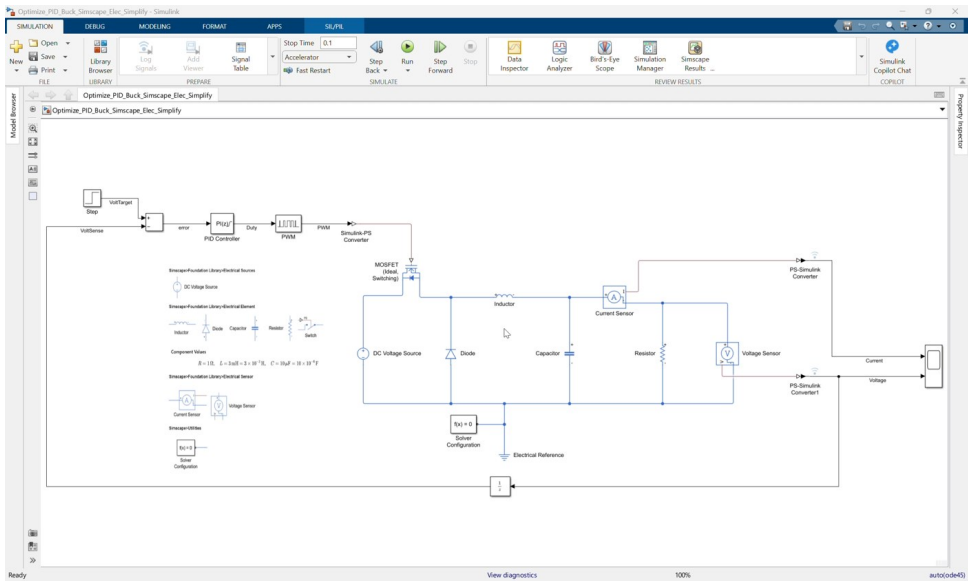
File	Purpose
Optimize_PID_Buck_Simscape_Elec_Simplify.slx	Starting optimization model

Finished Model Reference

File	Purpose
Optimize_PID_Buck_Simscape_Elec_Simplify_finished.slx	Finished optimized PI buck converter

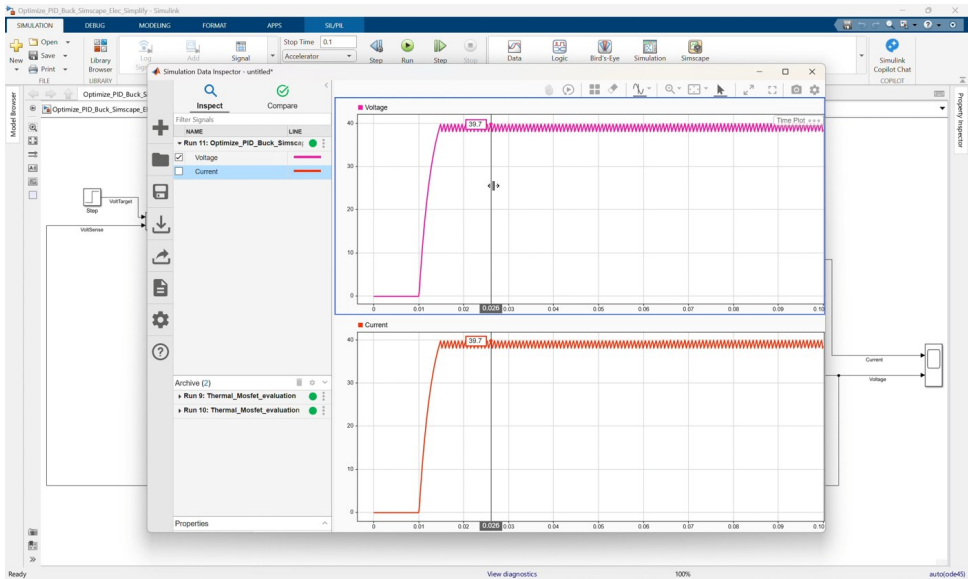
Workshop Procedure

1. Open and review the starting optimization model.



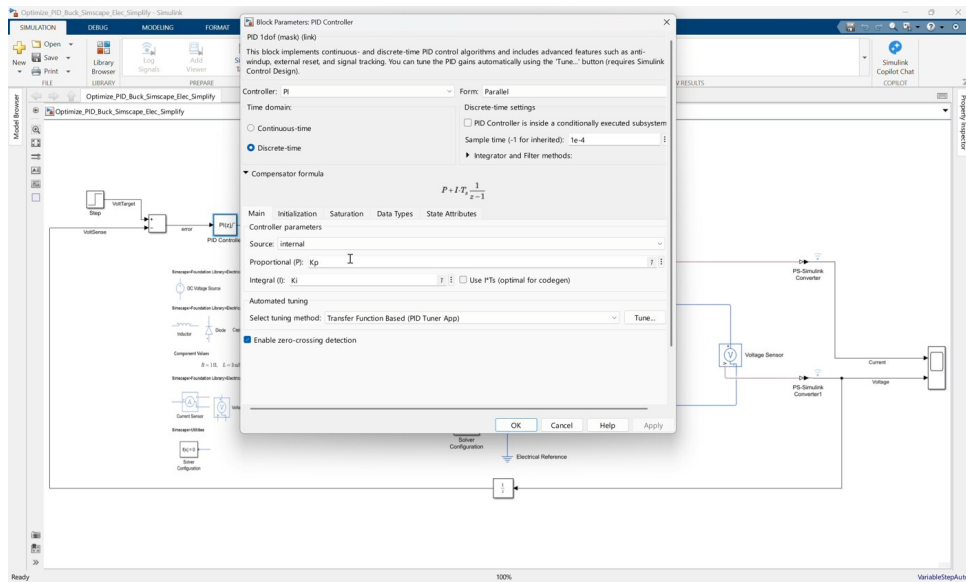
Starting optimization model

2. Review the step voltage target.



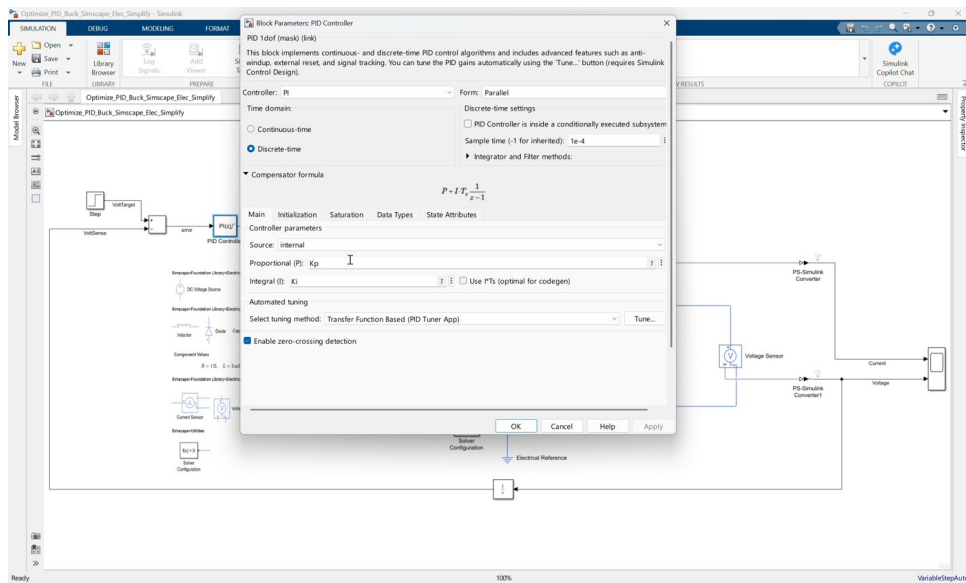
Step voltage target

3. Review the tunable PI controller using K_p and K_i .



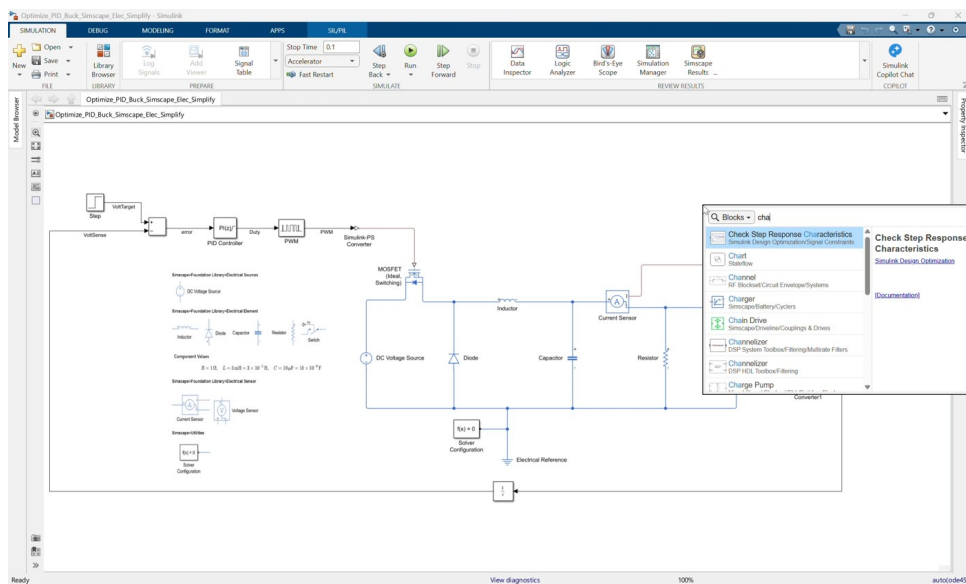
PI controller with Kp and Ki

4. Confirm workspace variables.



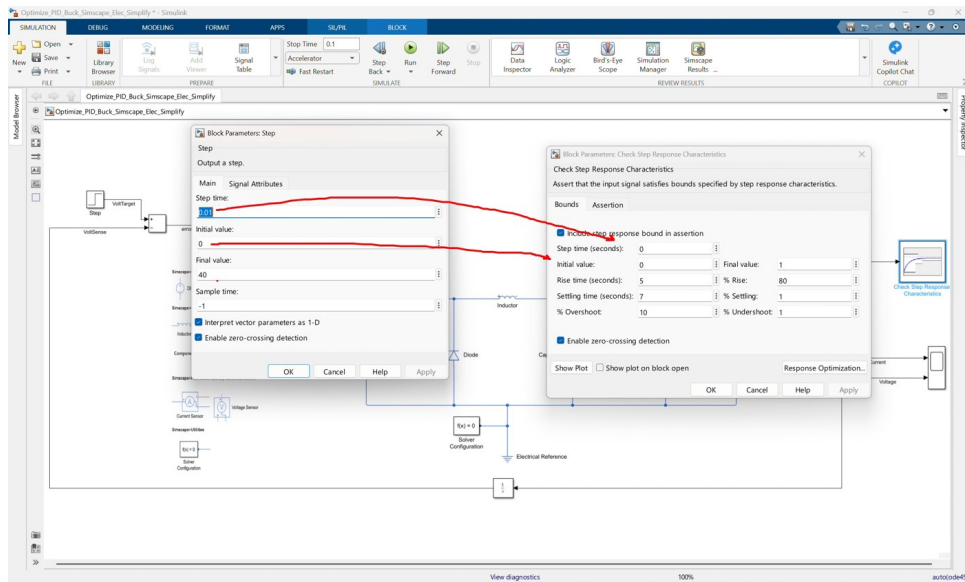
Workspace variables for optimization

5. Add the step response requirement block.



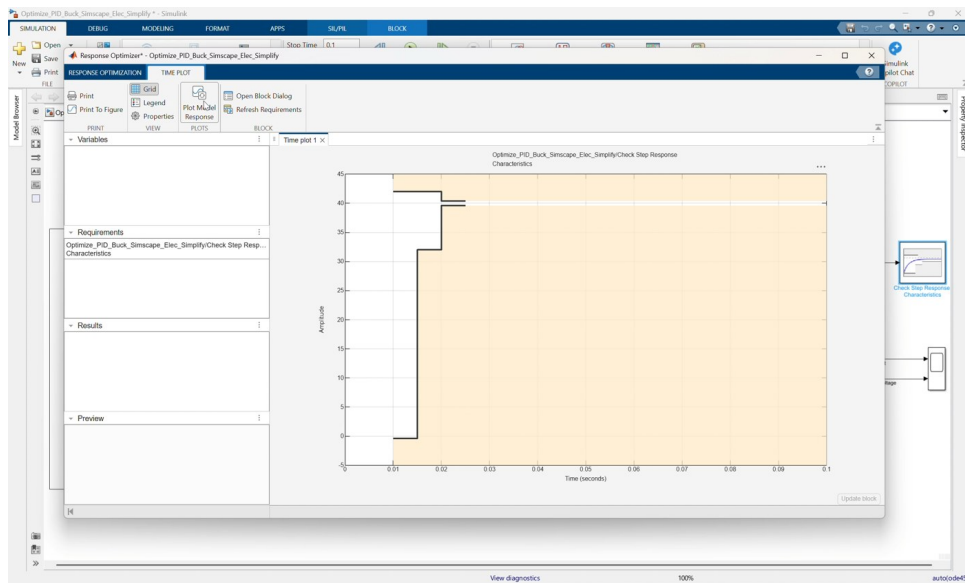
Add step response requirement

6. Configure rise time, settling time, overshoot, and target value.



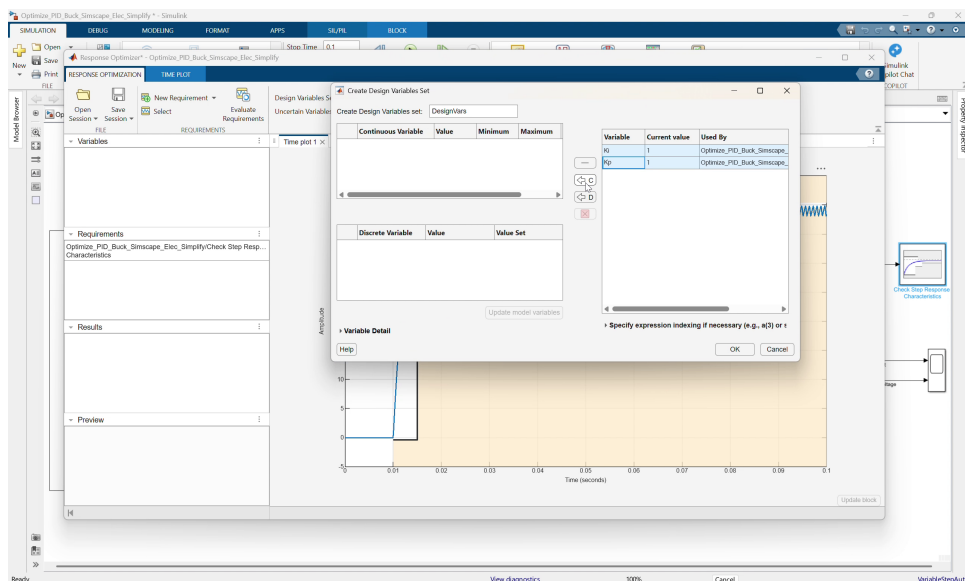
Configure step response requirement

7. Open Response Optimizer.



Open Response Optimizer

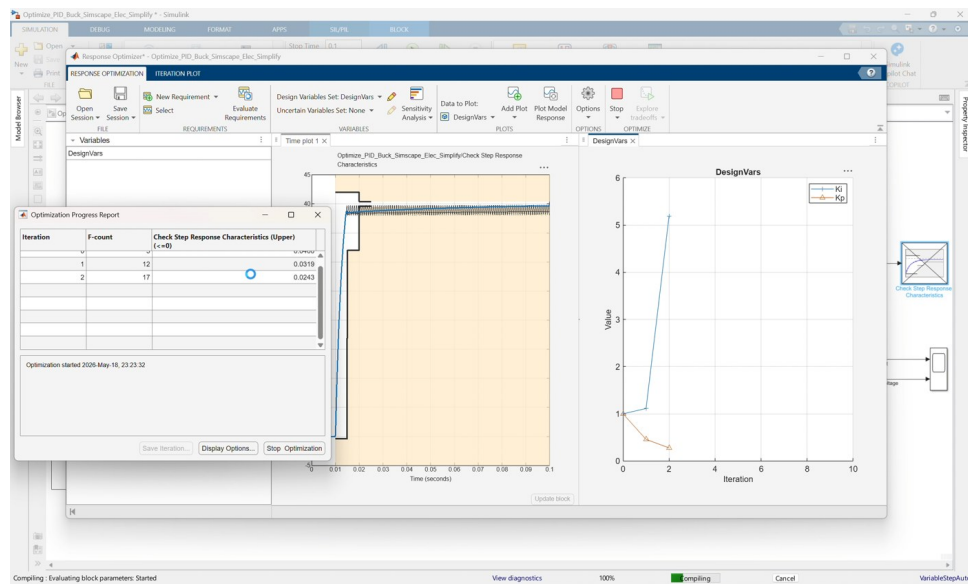
8. Add Kp and Ki as design variables.



Add design variables

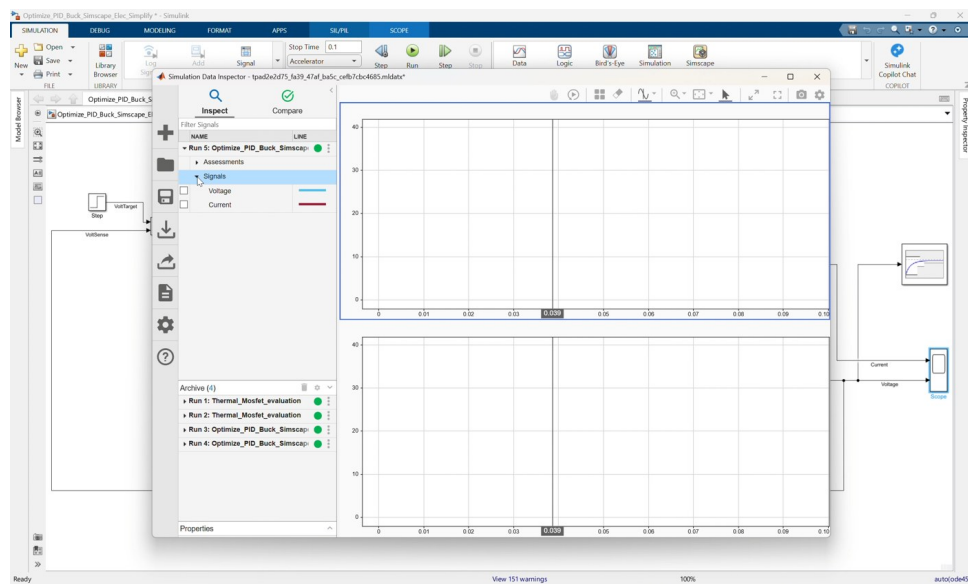
9. Configure simulation performance settings.

10. Run the optimization.



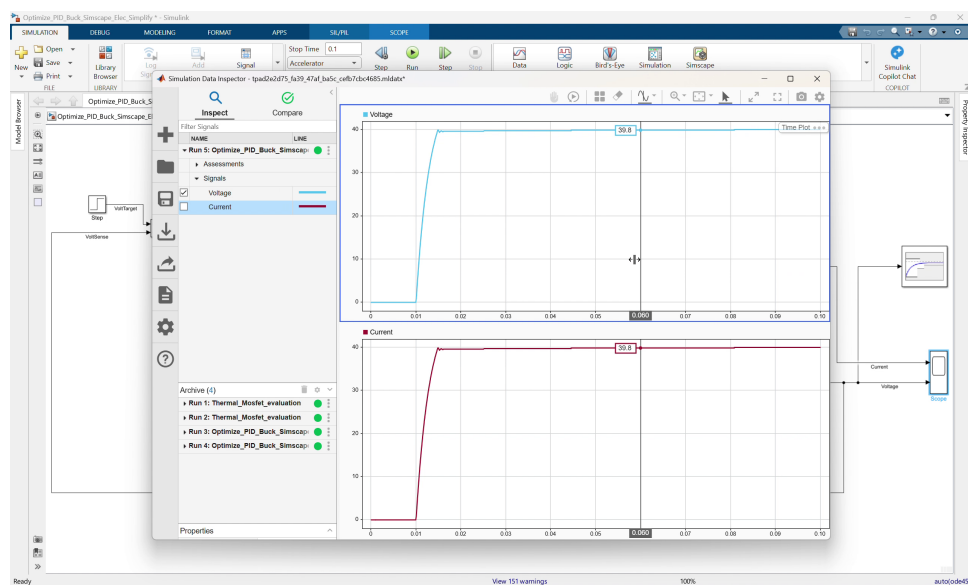
Run Response Optimizer

11. Review and apply the optimized values.



Review optimized gains

12. Validate the optimized response.



Validate optimized response

Workshop 05: Run SIL Simulation for the Buck Converter Controller

Starting Files

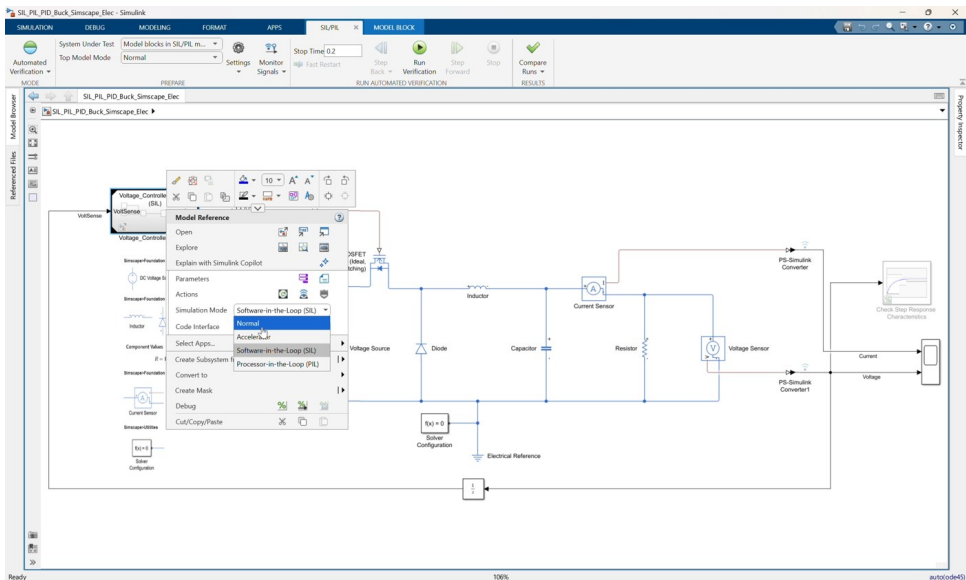
File or Folder	Purpose
SIL_PIL_PID_Buck_Simscape_Elec.slx	Top-level buck converter plant and controller reference model
Voltage_Controller_STM32.slx	Referenced controller model
Voltage_Controller_Project\	STM32 project configuration

Finished Model Reference

There is no separate finished model. This workshop demonstrates the SIL workflow using the prepared model and referenced controller.

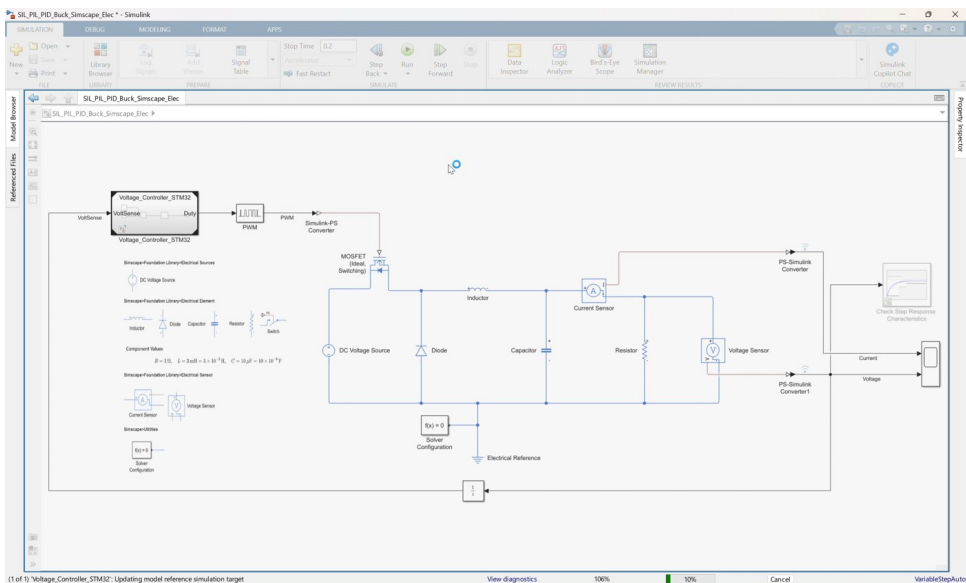
Workshop Procedure

1. Open the SIL/PIL top model.



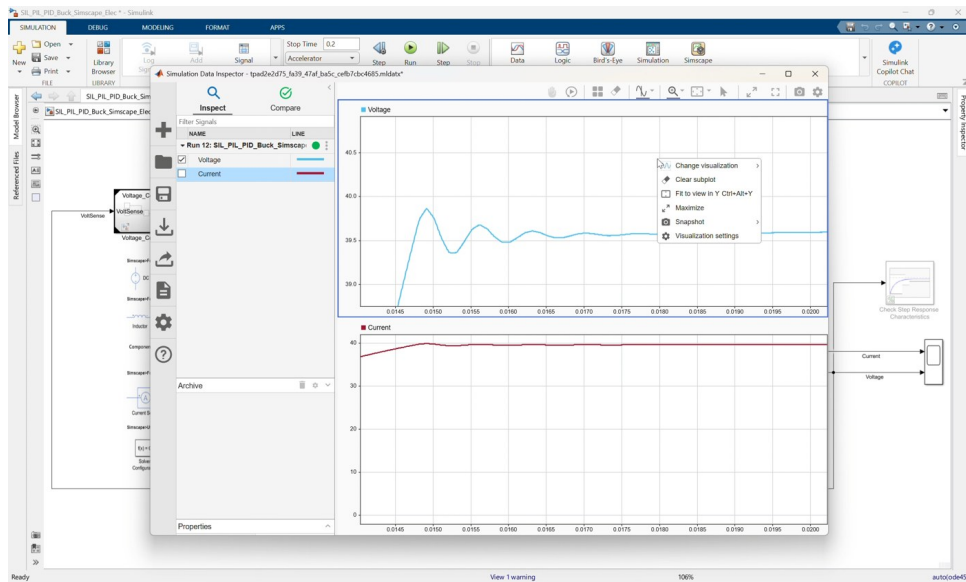
Open SIL/PIL top model

2. Review the closed-loop plant and controller boundary.



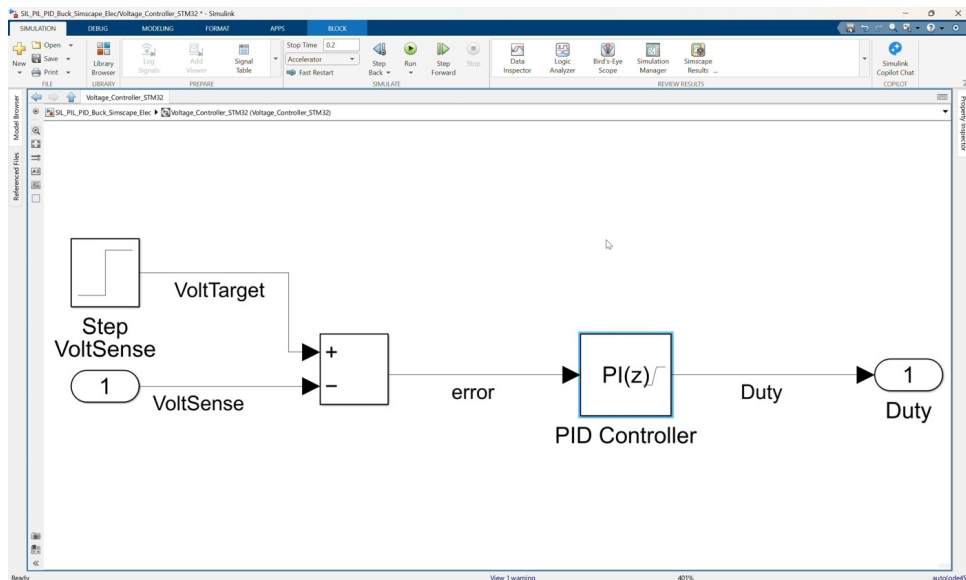
Controller boundary

3. Run a baseline model simulation.



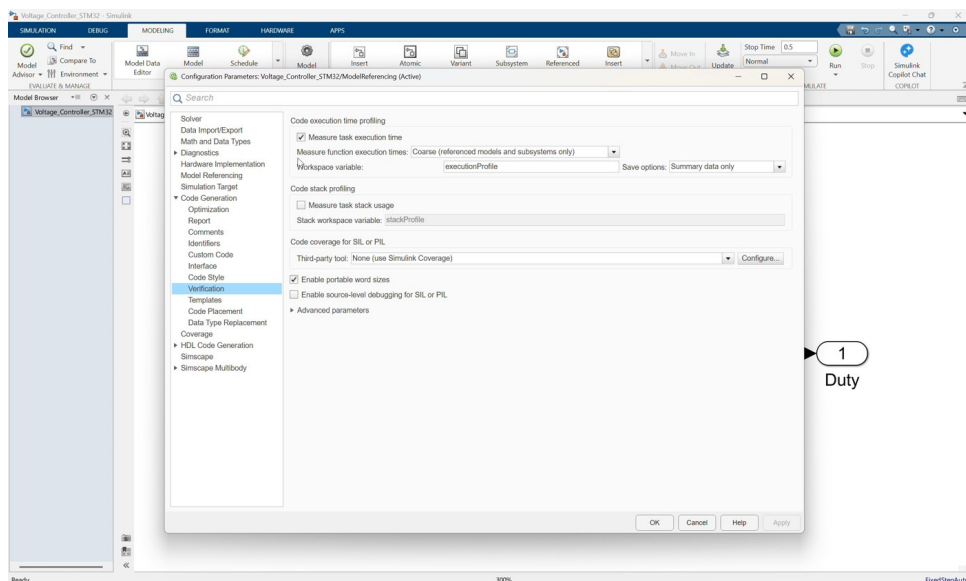
Baseline model simulation

4. Open the Voltage_Controller_STM32 reference model.



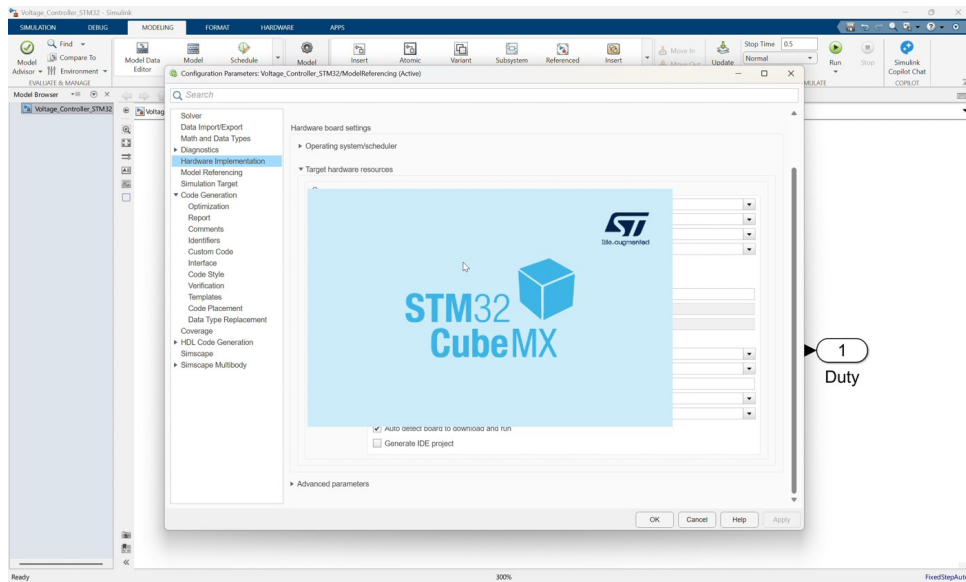
Controller reference model

5. Review code generation settings.



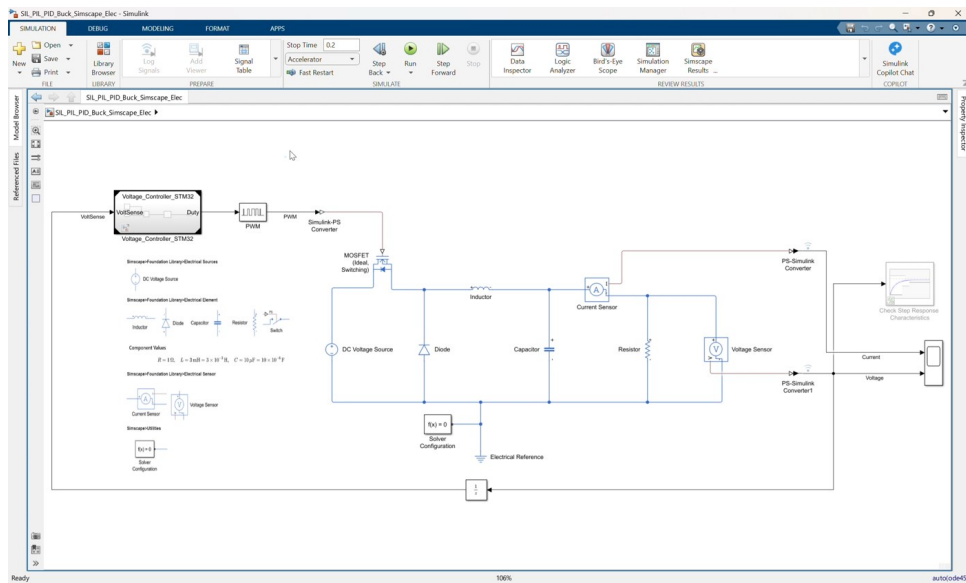
Code generation settings

6. Review the STM32 project configuration.



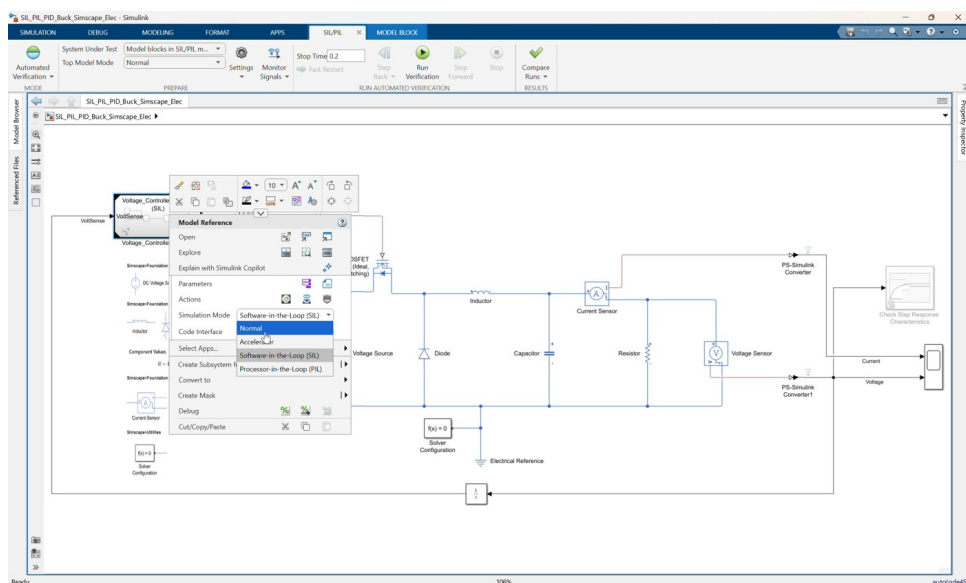
STM32 project configuration

7. Build or confirm generated code.



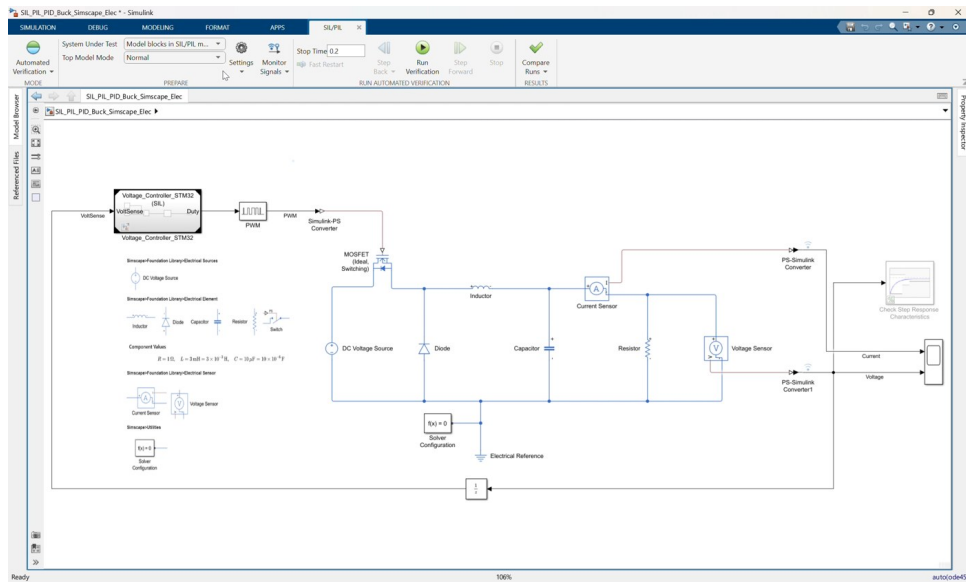
Generated code artifacts

8. Switch the Model Reference block to Software-in-the-loop (SIL).



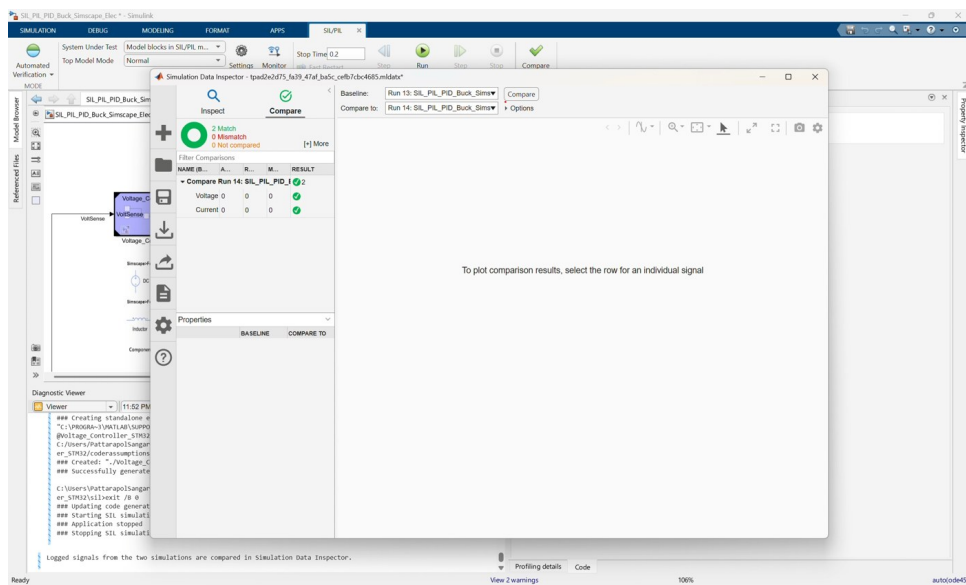
Switch to SIL mode

9. Run SIL simulation.



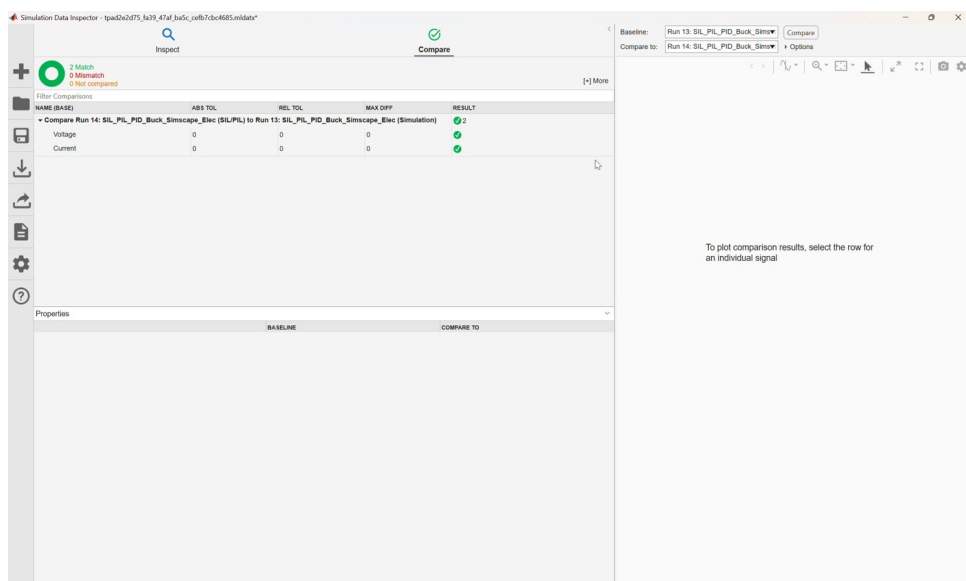
Run SIL simulation

10. Compare SIL result with model simulation.



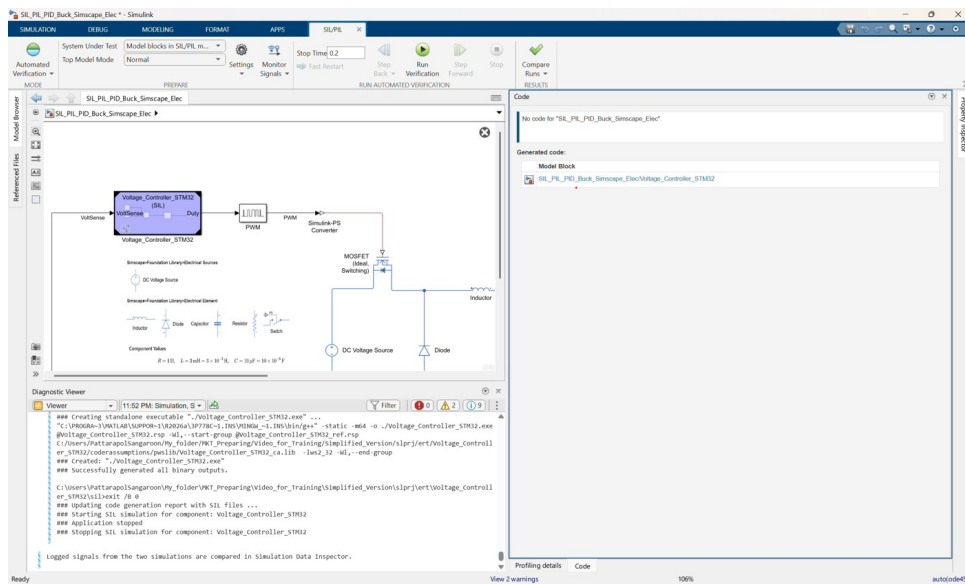
Compare SIL results

11. Open the generated code report.



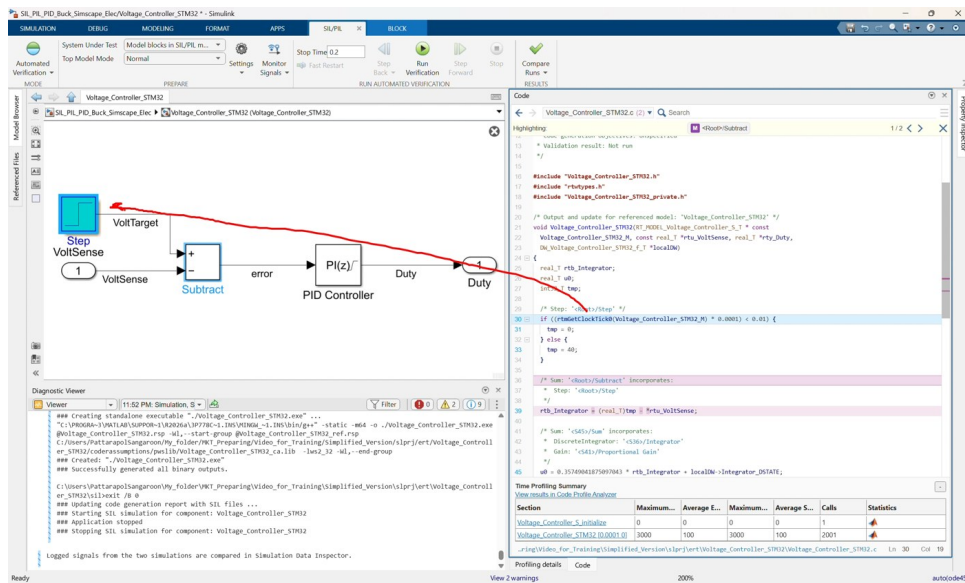
Generated code report

12. Demonstrate model-to-code traceability.



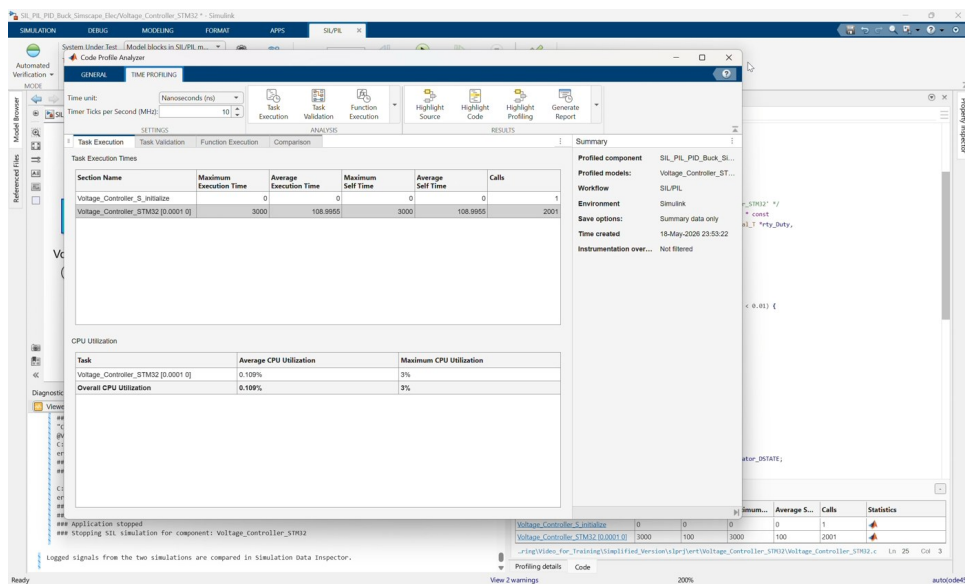
Model-to-code traceability

13. Demonstrate code-to-model traceability.



Code-to-model traceability

14. Inspect code execution profiling.



Code execution profiling

Workshop 06: Run PIL Simulation for the Buck Converter Controller

Starting Files

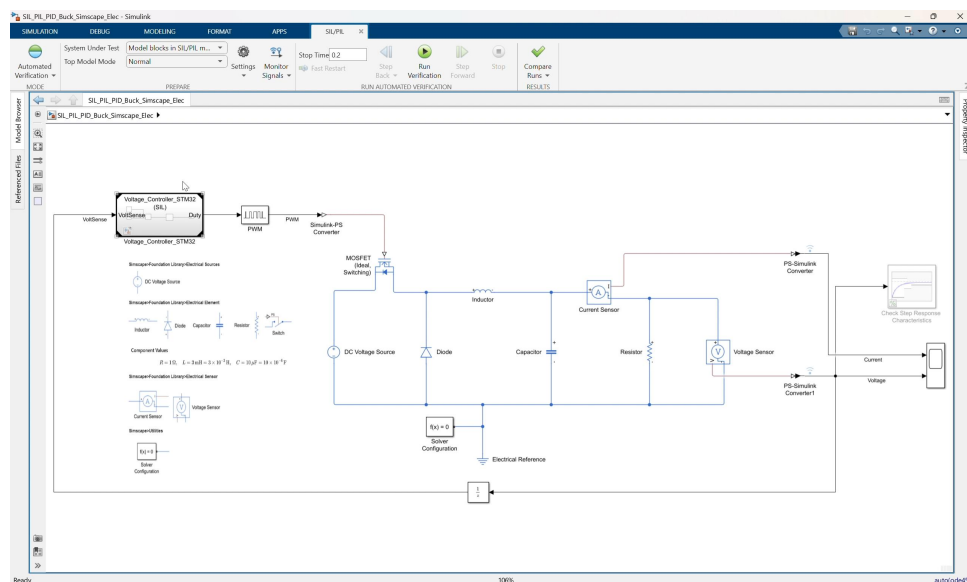
File or Folder	Purpose
SIL_PIL_PID_Buck_Simscape_Elec.slx	Top-level buck converter plant and controller reference model
Voltage_Controller_STM32.slx	Referenced controller model
Voltage_Controller_Project\	STM32 project configuration

Finished Model Reference

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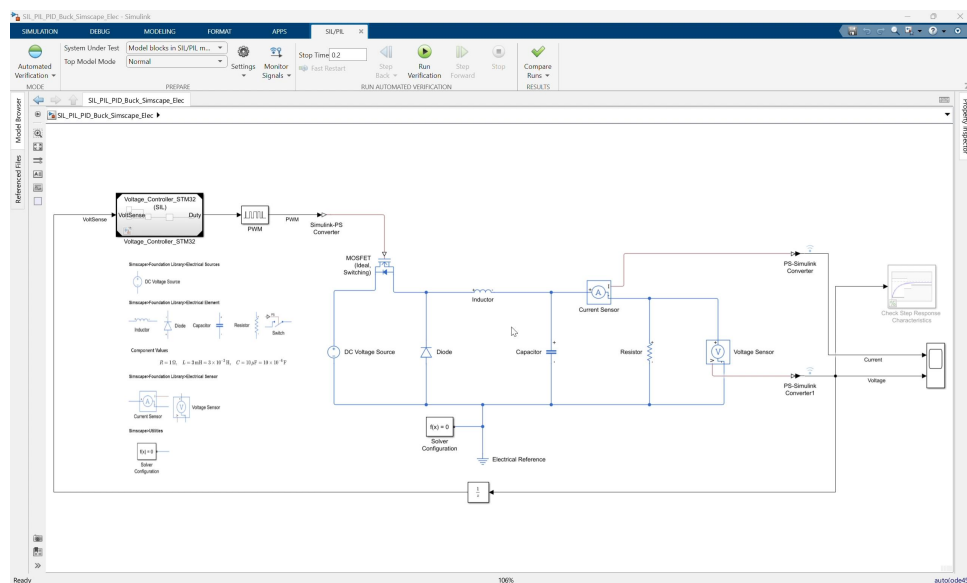
Workshop Procedure

1. Open the SIL/PIL top model.



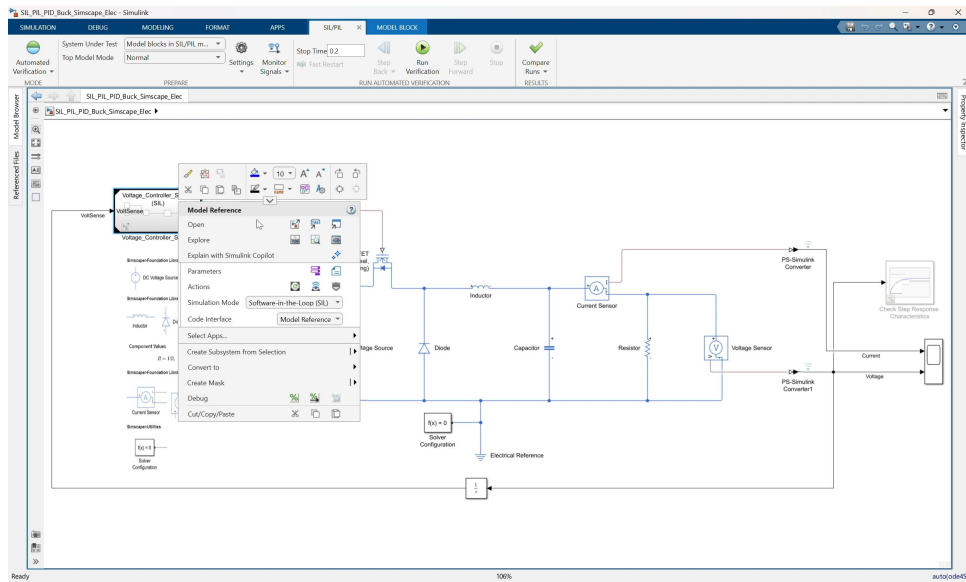
Open SIL/PIL top model

2. Review the closed-loop plant and controller boundary.



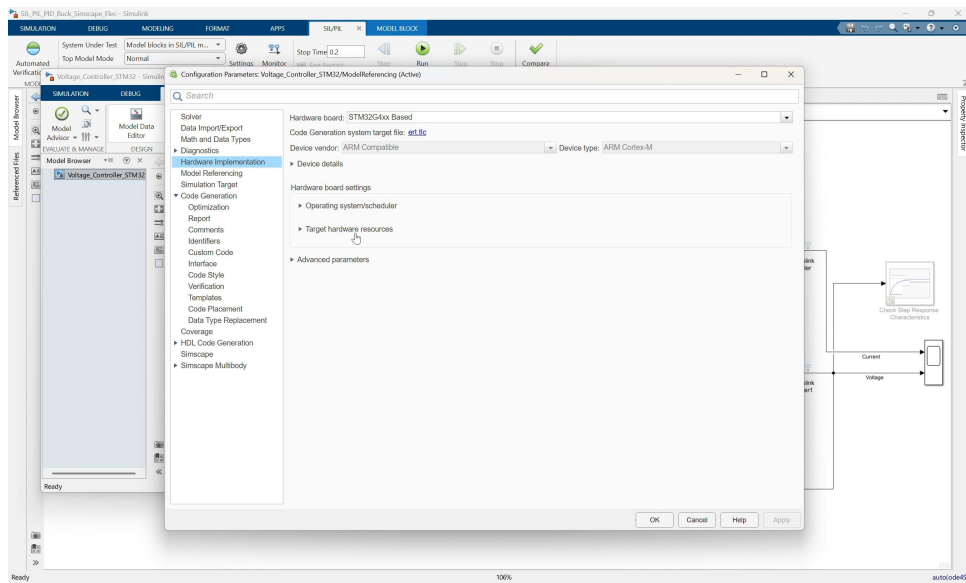
Controller boundary

3. Select Processor-in-the-loop (PIL) mode.



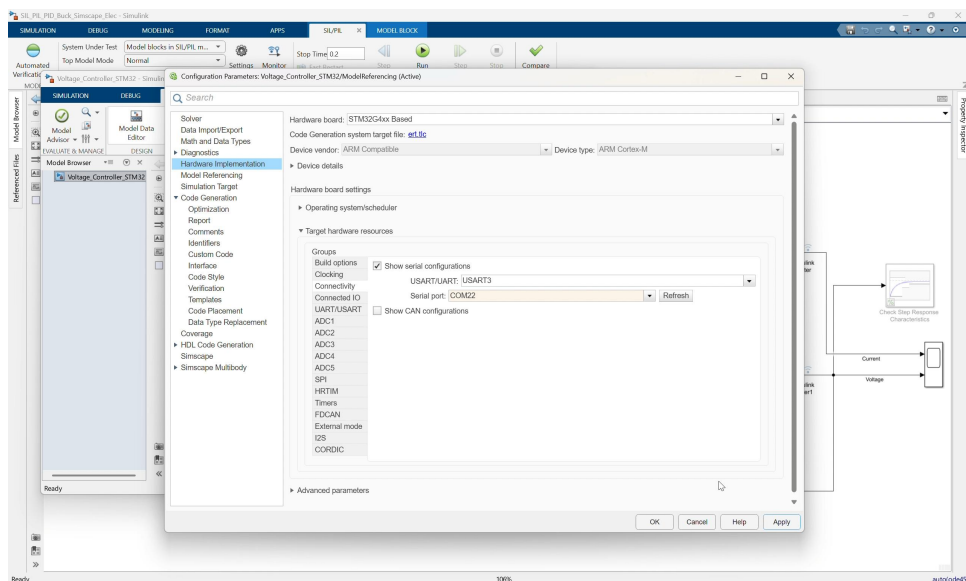
Select PIL mode

4. Open the controller model settings.



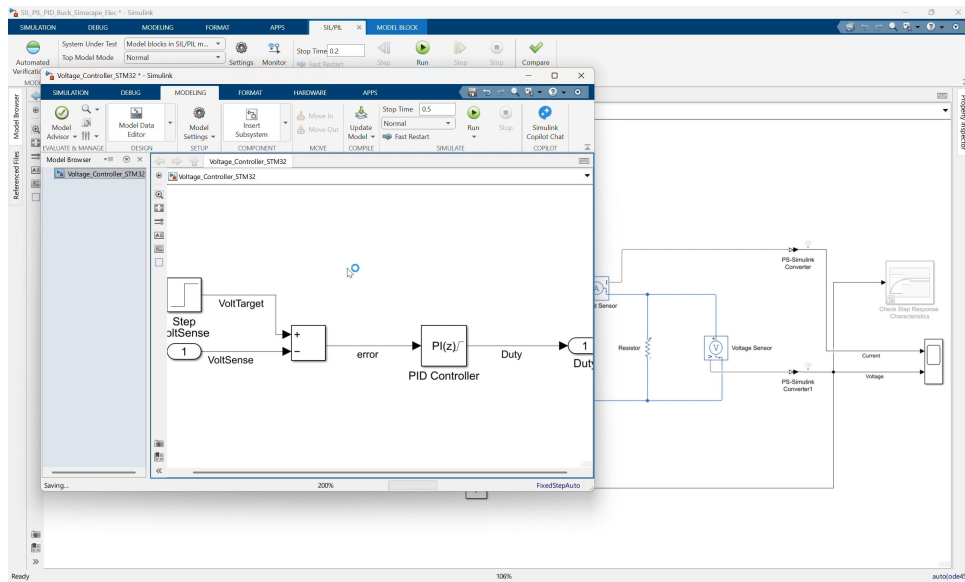
Controller model settings

5. Review target hardware resources and communication settings.



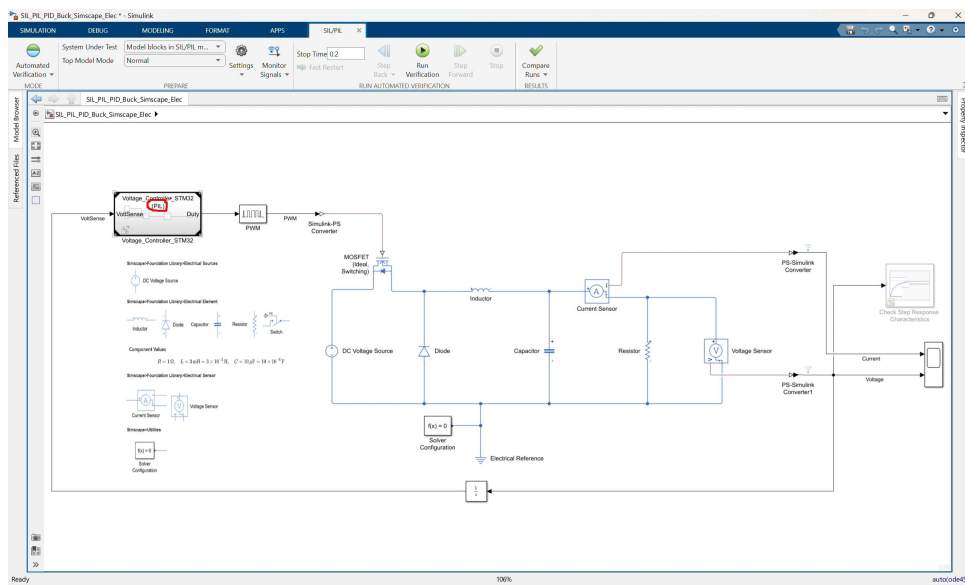
Target hardware resources

6. Review the referenced controller logic.



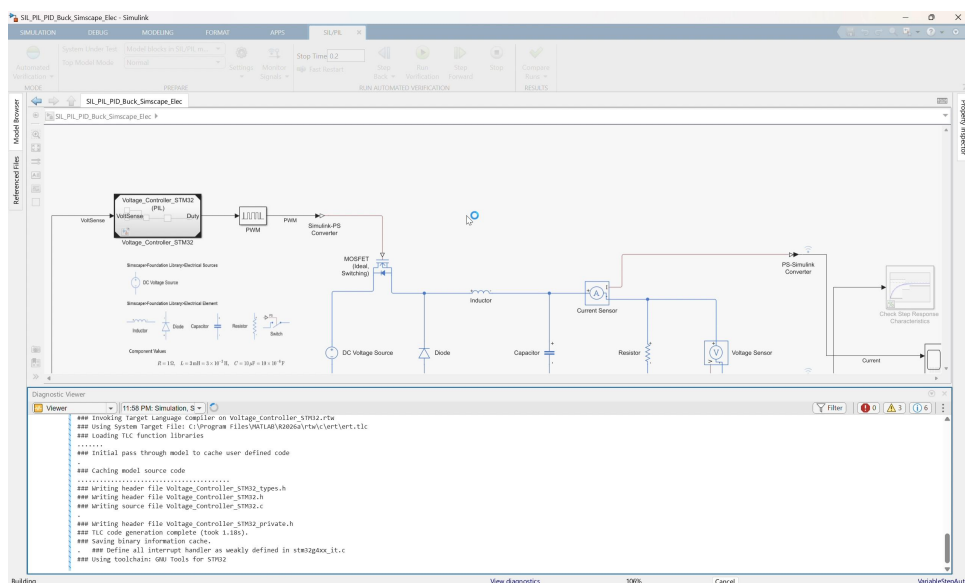
Referenced controller logic

7. Confirm generated code and PIL artifacts.



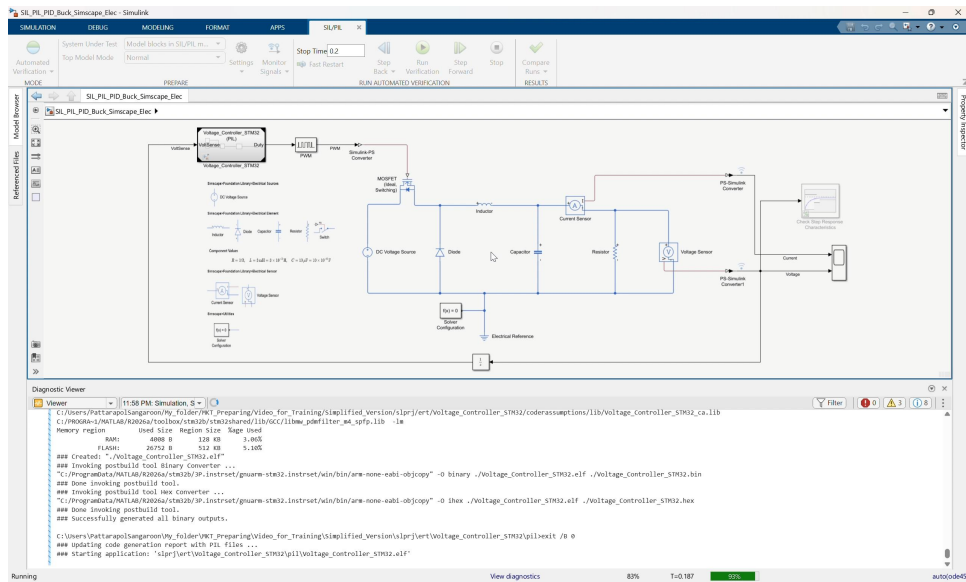
PIL mode confirmed

8. Run the PIL simulation.



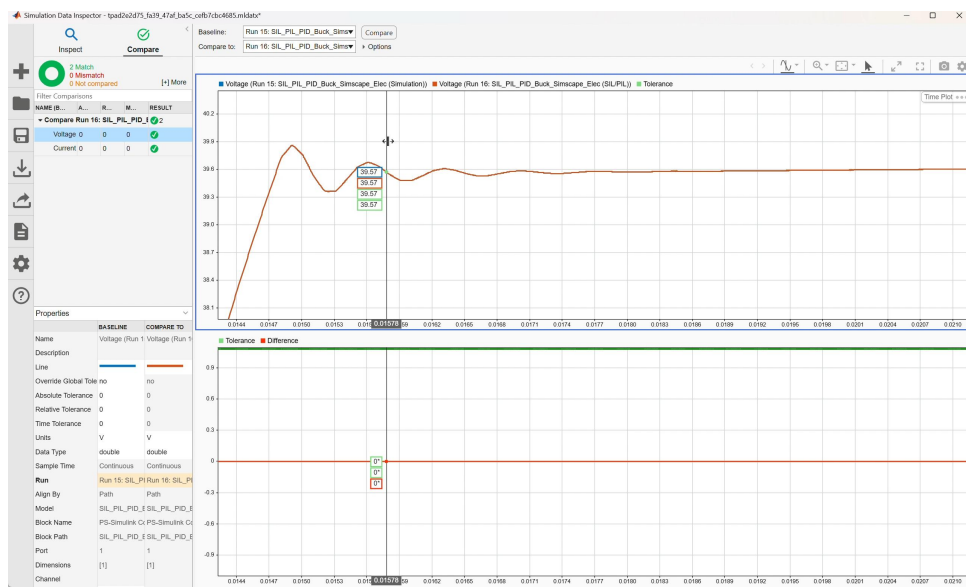
Run PIL build

9. Wait for build, deploy, and target execution.



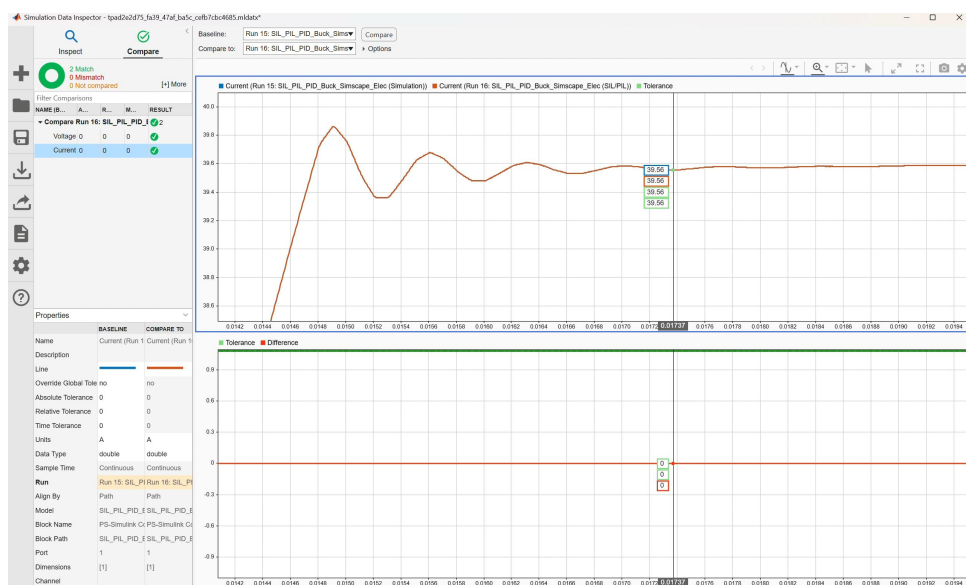
Target execution

10. Compare simulation and PIL voltage results.



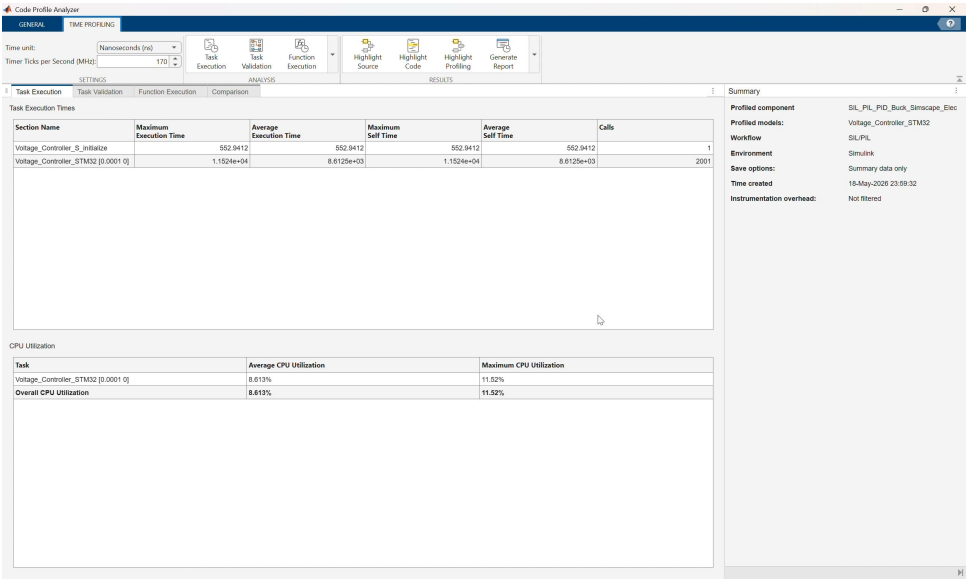
Compare voltage results

11. Compare current results and open Code Profile Analyzer.



Compare current results

12. Interpret task execution time and CPU utilization.



Code Profile Analyzer